自测题

《有机化学》

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科学出版社

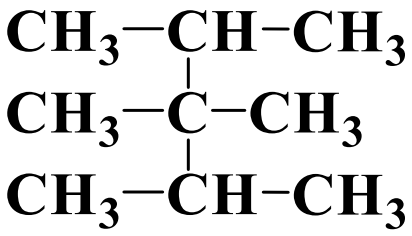
第2章　饱和烃

1. 命名

(1). $(CH_3)_2CHCH(CH_3)_2$

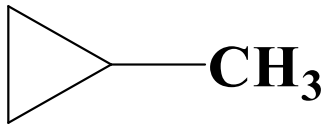
2,3-二甲基丁烷

(2).



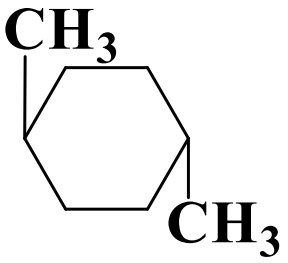
2,3,3,4-四甲基戊烷

(3).



甲基环丙烷

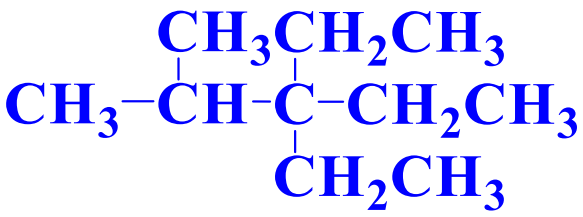
(4).



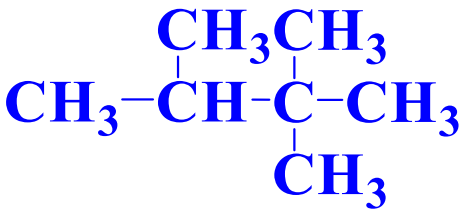
反-1,4-二甲基环己烷

2. 写结构式

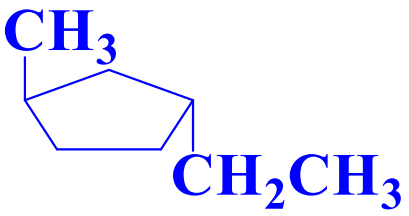
(1). 2-甲基-3,3-二乙基戊烷



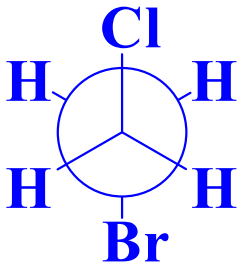
(2). 一个异丙基和一个叔丁基组成的烷烃



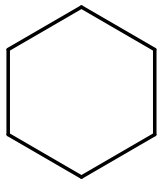
(3). 反-1-甲基-3-乙基环戊烷

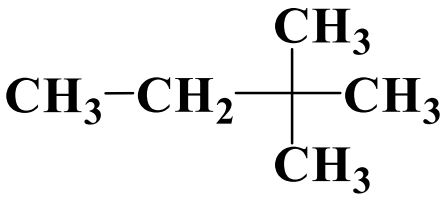


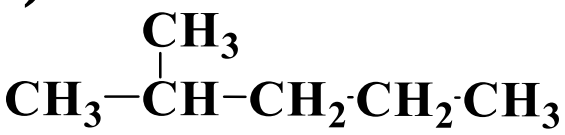
(4). 1-氯-2-溴乙烷（优势构象）



3. 不查表比较下列化合物沸点的高低

(1). 

(2). 

(3). 

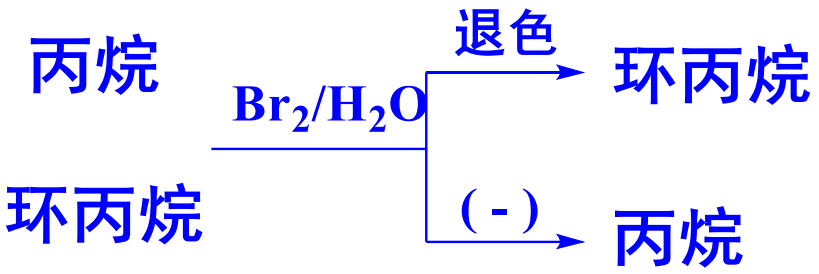
(4). $CH_3CH_2CH_2CH_2CH_2CH_3$

(1) > (4) > (3) > (2)

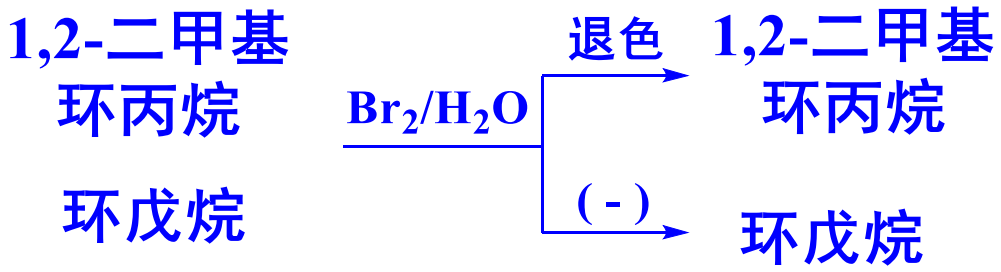
4. 用简单的化学方法区别下列各组化合物

(1) 丙烷和环丙烷

分别加入溴水溶液，退色的是环丙烷，不变的是丙烷

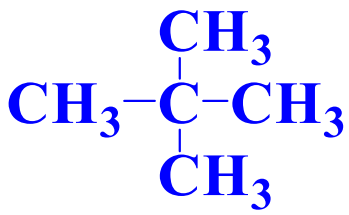


(2) 1,2-二甲基环丙烷和环戊烷



5. 推导结构式:

化合物A、B、C是互为同分异构体的烷烃, 相对分子质量均为72。A中仅含有伯碳和季碳，B中含有伯、仲、叔碳, C中仅含有伯、仲碳原子。写出A、B、C的结构式。

A: 

B: 

C: $CH_3CH_2CH_2CH_2CH_3$

第3章 不饱和烃

1. 命名：

(1) $CH_2{=}CH{-}CH_2{-}$

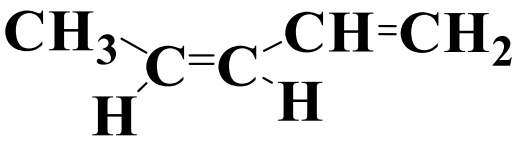
烯丙基

(2) $CH_3{-}CH{=}CH{-}$

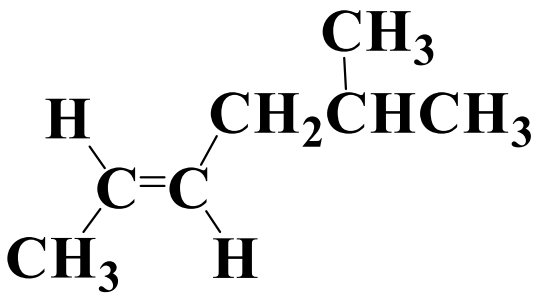
丙烯基

(3) $CH_3CH_2{-}\underset{\underset{CH_2}{\|}}{C}{-}CH(CH_3)_2$

3-甲基-2-乙基-1-丁烯

(4) 

顺-1,3-戊二烯

(5) 

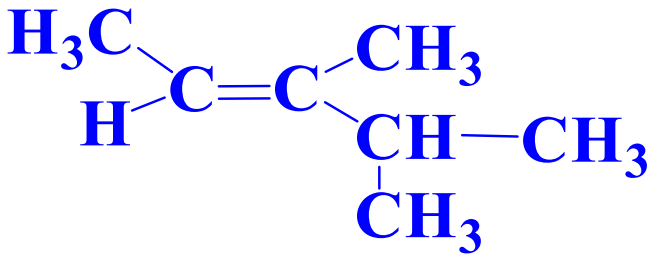
反-5-甲基-2-己烯

(6) $HC{\equiv}C{-}\underset{\underset{CH_3}{|}}{CH}CH_3$

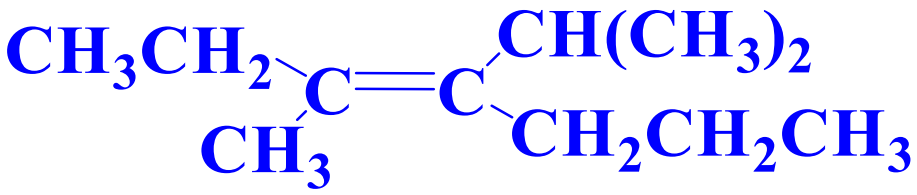
3-甲基-1-丁炔

2. 写结构式：

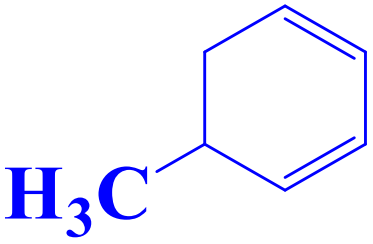
(1). E-3,4-二甲基-2-戊烯



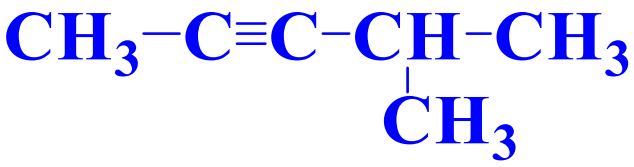
(2). Z-3-甲基-4-异丙基-3-庚烯



(3). 5-甲基-1,3-环己二烯



(4). 4-甲基-2-戊炔



3、选择题：

(1). 下列碳正离子最稳定的是（ A ）

A. $(CH_3)_2C{=}CH\overset{+}{C}H_2$　　B. $CH_3CH{=}CH{-}\overset{+}{C}H_2$

C. $CH_2{=}CH{-}\overset{+}{C}H_2$　　D. $CH_3CH_2CH_2\overset{+}{C}H_2$

(2). 下列化合物与HBr反应时，速度最快的是（ D），最慢的是（ A ）

A. $CH_2{=}CH_2$　　B. $CH_3CH{=}CH_2$　　C. $CH_3CH{=}CHCH_3$　　D. $(CH_3)_2C{=}CHCH_3$

(3). 下列化合物中有顺反异构体的是（ D ）

A. 2-甲基-1-丁烯；　　B. 丙烯；

C. 3-甲基-4-乙基-3-己烯；　D. 2,5-二甲基-3-己烯

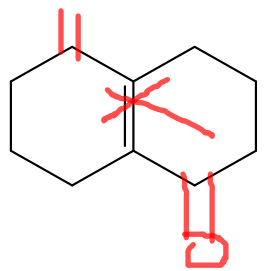
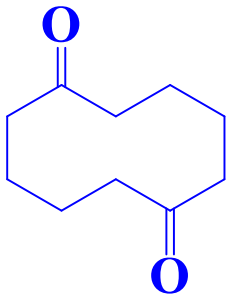
4. 完成下列反应式

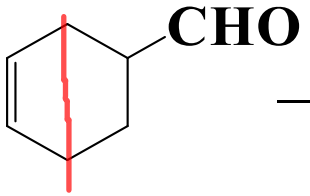
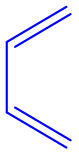
(1). $(CH_3)_2C{=}CH_2 \xrightarrow{\text{冷、稀 } KMnO_4}$ ($H_3C-\underset{OH}{\overset{CH_3}{C}}-\underset{OH}{CH_2}$)

(2). $CH_3CH_2-\underset{CH_3}{C}{=}CH_2 + HBr \longrightarrow$ ($CH_3CH_2-\overset{Br}{\underset{CH_3}{C}}-CH_3$)

(3). $CH_3CH_2-\underset{CH_3}{C}{=}CH_2 + HBr \xrightarrow{\text{过氧化物}}$ ($CH_3CH_2-\overset{H}{\underset{CH_3}{C}}-CH_2Br$)

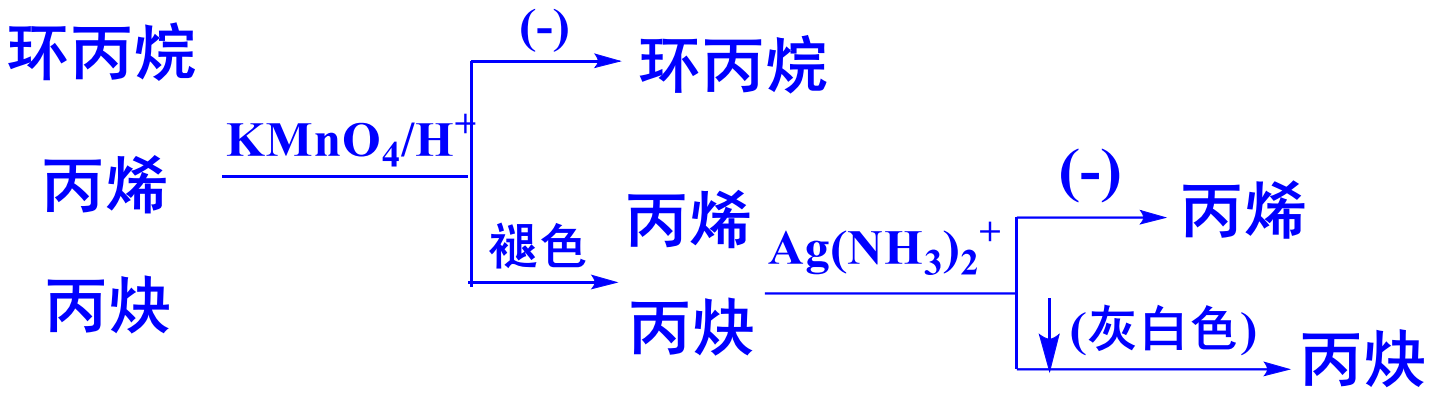
(4). $(CH_3)_2C{=}CH-CH{=}CH_2 + HBr \longrightarrow$ ($(CH_3)_2\underset{Br}{CH}-CH{=}CH-CH_3$)

(5).  $\xrightarrow[(2)\ Zn,\ H_2O]{(1)\ O_3}$ ()

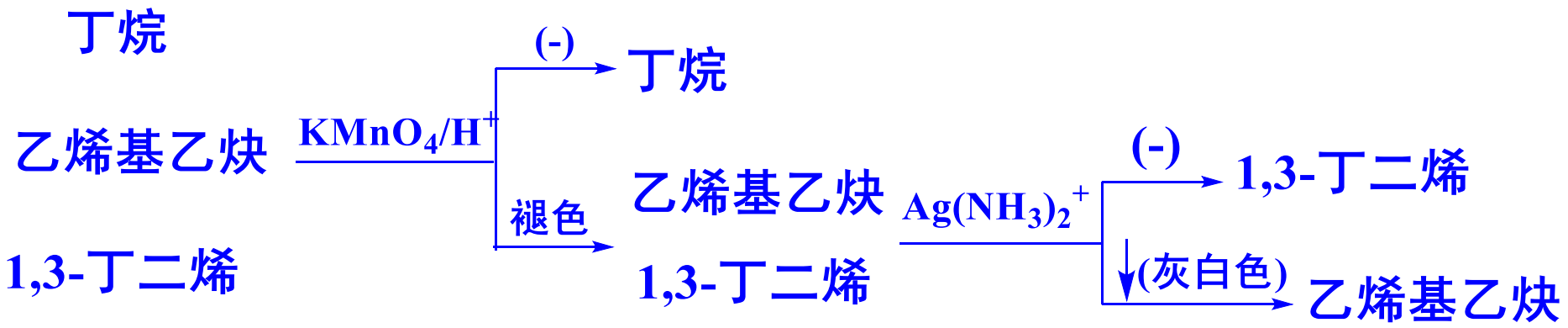
(6).  $\xrightarrow{\Delta}$ () + ()

5. 用简单的化学方法区别下列各组化合物：

(1). 环丙烷　　丙烯　　丙炔



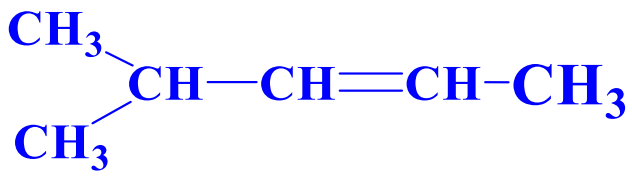
(2). 丁烷　　乙烯基乙炔　　1,3-丁二烯



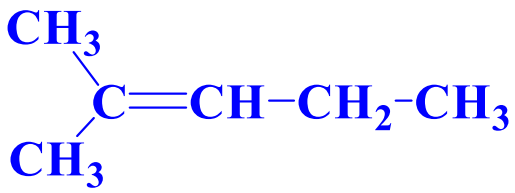
6. 推导结构式

(1). 有两种烯烃A 和B，经催化加氢都得到C。A用$KMnO_4/H^+$氧化，得到CH_3COOH和$(CH_3)_2CHCOOH$；B在同样条件下得到丙酮和CH_3CH_2COOH。写出A、B、C的结构式

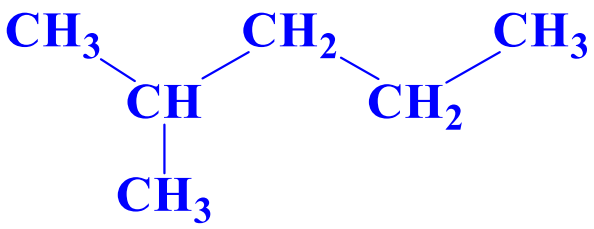
A.



B.



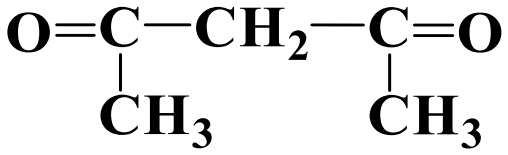
C.

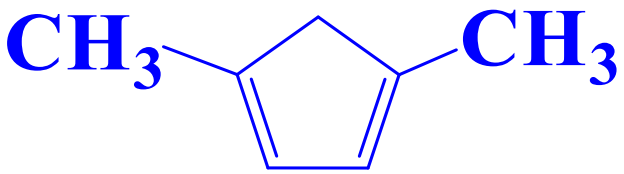


(2). 化合物A和B的分子式同为C_4H_6，都能使溴的四氯化碳溶液褪色。A与硝酸银的氨溶液作用产生沉淀，氧化A得CO_2和丙酸，B不与硝酸银的氨溶液作用，氧化B得CO_2和草酸(HOOC-COOH)。试推测A与B的构造式

A. $CH_3CH_2C{\equiv}CH$

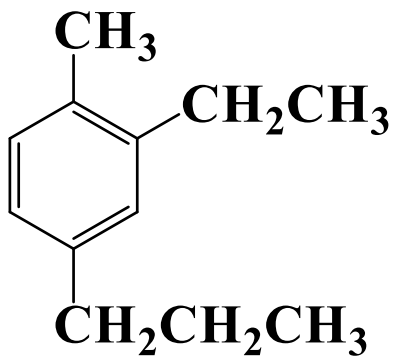
B. $CH_2{=}CH{-}CH{=}CH_2$

(3). 某烃A分子式为C_7H_{10}，用高锰酸钾氧化后得到草酸(HOOC-COOH)及  ，写出此烃A的结构式



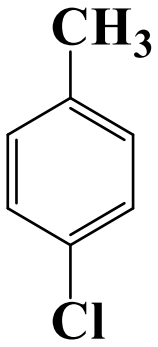
1. 命名

(1)



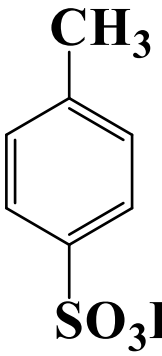
2-乙基-4-正丙基甲苯

(2)



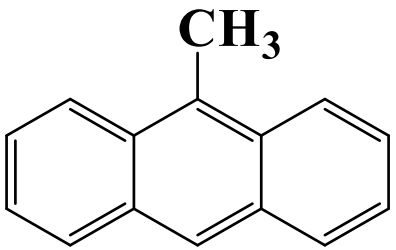
4-氯甲苯

(3)



4-甲基苯磺酸

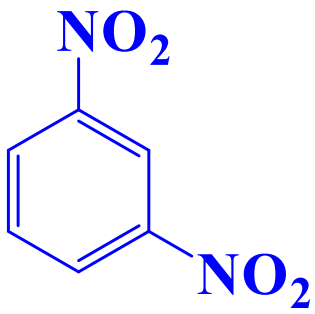
(4)



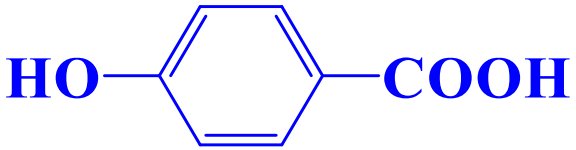
9-甲基蒽

2. 写出下列化合物的结构式：

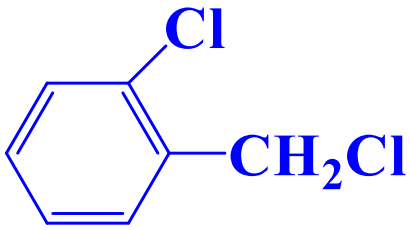
(1). 间二硝基苯



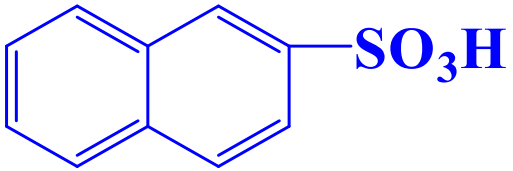
(2). 对羟基苯甲酸



(3). 邻氯苄氯



(4) . β –萘磺酸



3. 判断题

(1). 苯比氯苯更容易发生亲电取代反应. （ √ ）

(2). 苯分子中有3个碳碳双键，所以苯易与HBr发生亲电加成反应 （ × ）

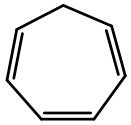
(3). 异丙苯在光照条件下发生氯代反应，其反应历程是亲电取代 （ × ）

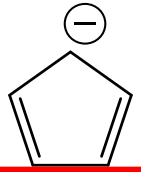
(4). 环丙烯正离子比三级碳正离子稳定 （ √ ）

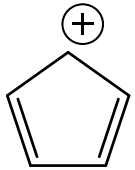
(5). 苯、萘、蒽、菲及环丁二烯都有芳香性 （ × ）

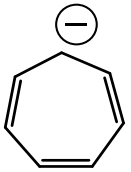
4. 选择题

(1). 下列结构中有芳香性的是(B.)

A. 

B. 

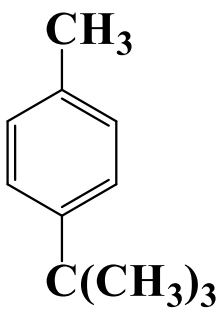
C. 

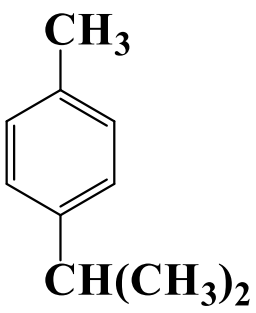
D. 

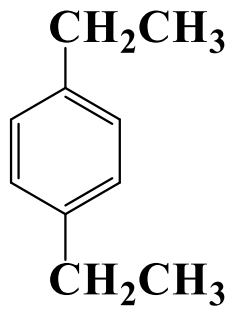
(2). 下列化合物中最易发生亲电取代反应的是(D.)

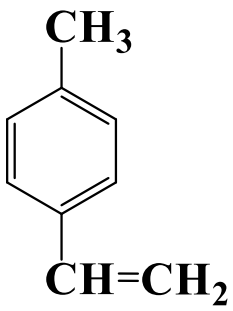
A. 苯　　B. 硝基苯　　C. 甲苯　　D. 苯酚

(3). 下列化合物用高锰酸钾氧化能生成一元芳香 (A.)

A. 

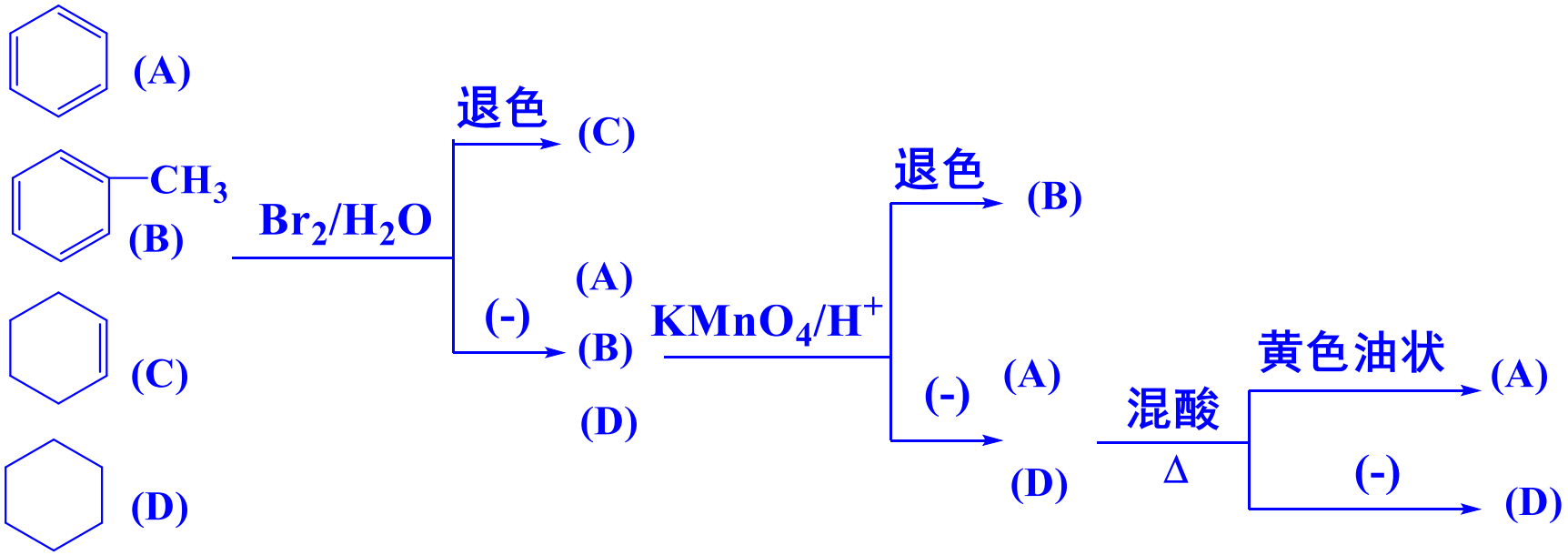
B. 

C. 

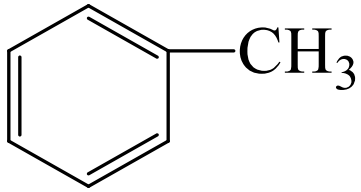
D. 

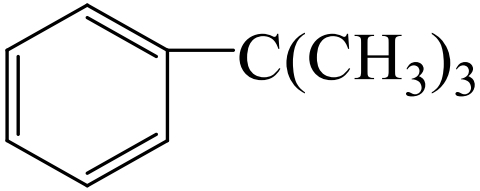
5. 用简单的化学方法鉴别下列各组化合物

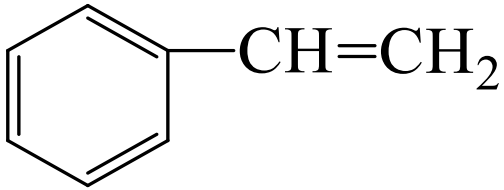
(1). 苯　　甲苯　　环己烯　　环己烷

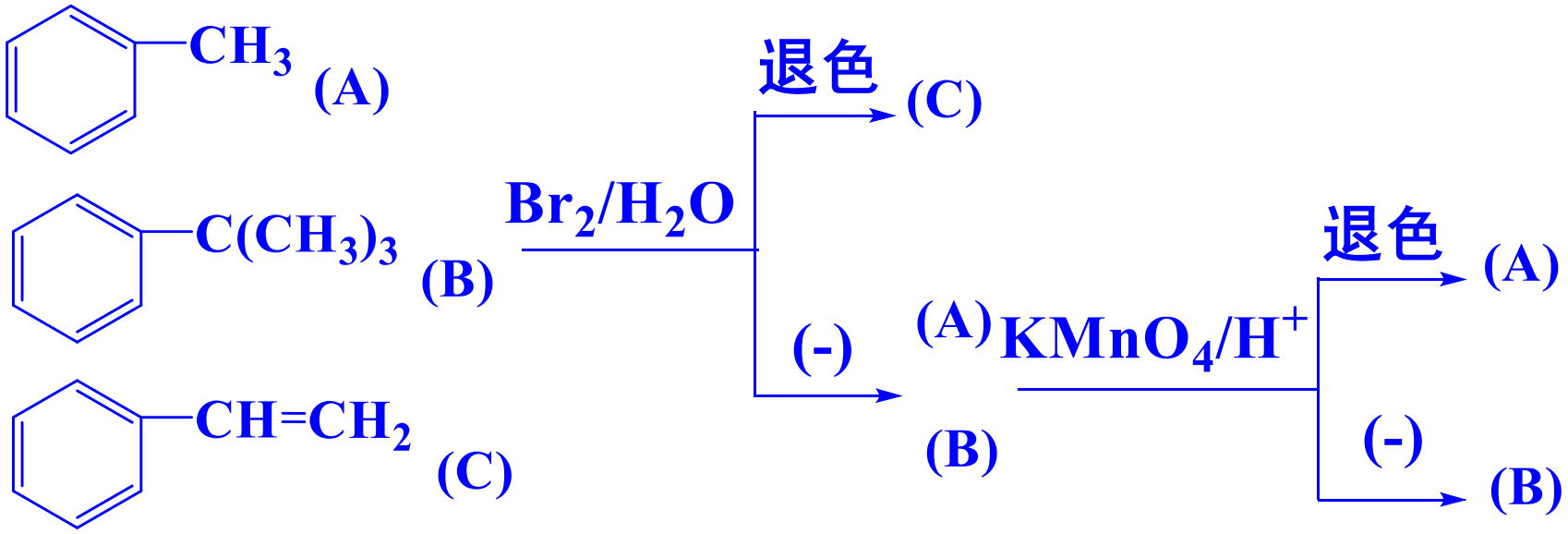


(2).


CH3

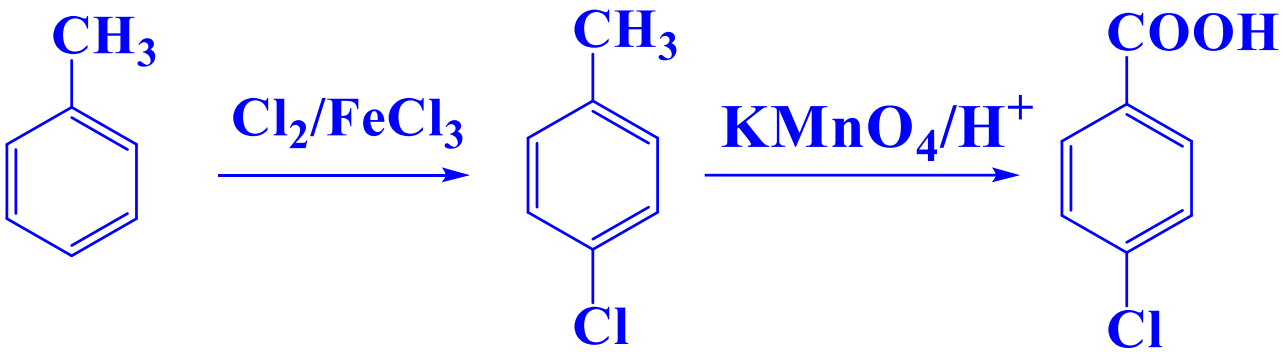

C(CH3)3


CH=CH2

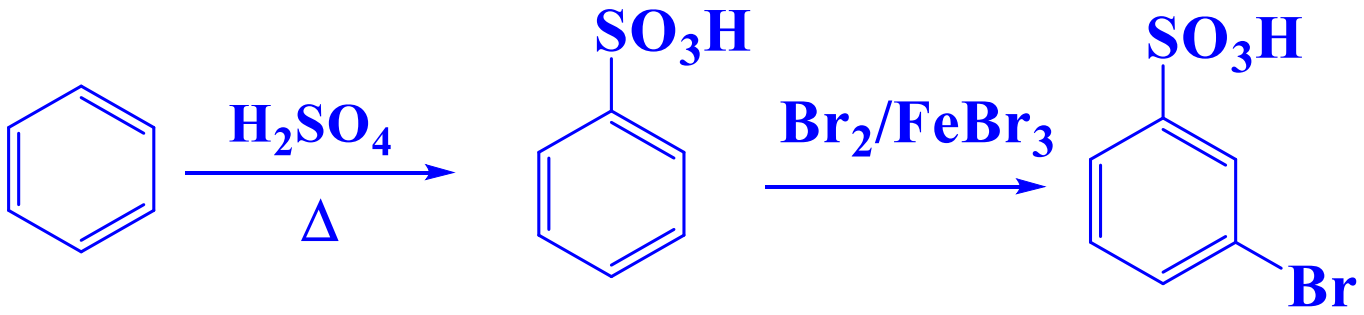

CH3 (A)
C(CH3)3 (B)
CH=CH2 (C)
Br2/H2O
退色
(C)
(-)
(A)
(B)
KMnO4/H+
退色
(A)
(-)
(B)

6. 用苯或甲苯为原料合成化合物

(1). 对氯苯甲酸

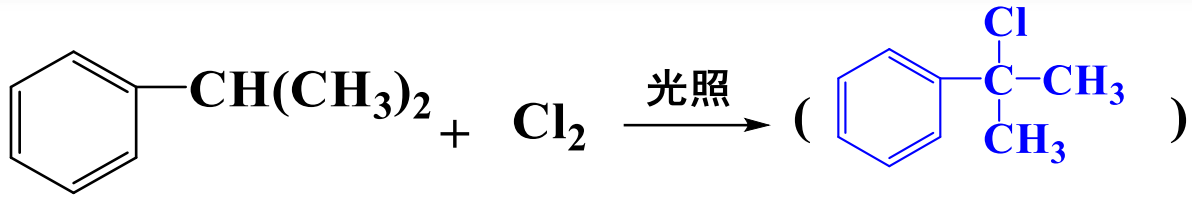


(2). 间溴苯磺酸

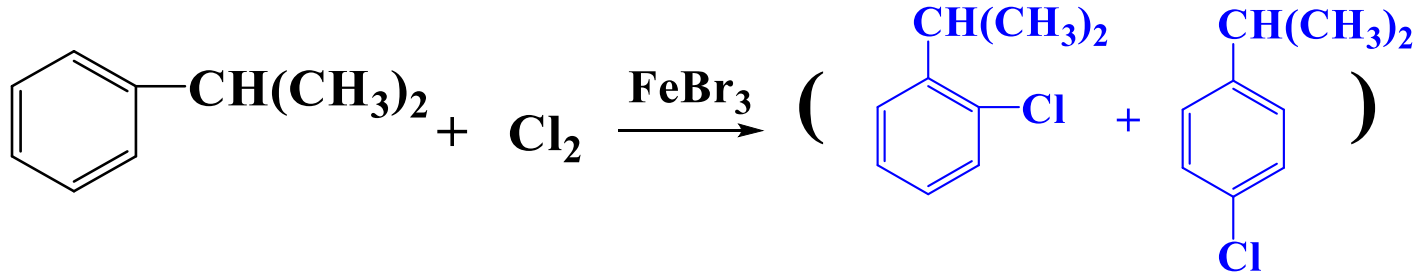


7. 完成下列反应式：

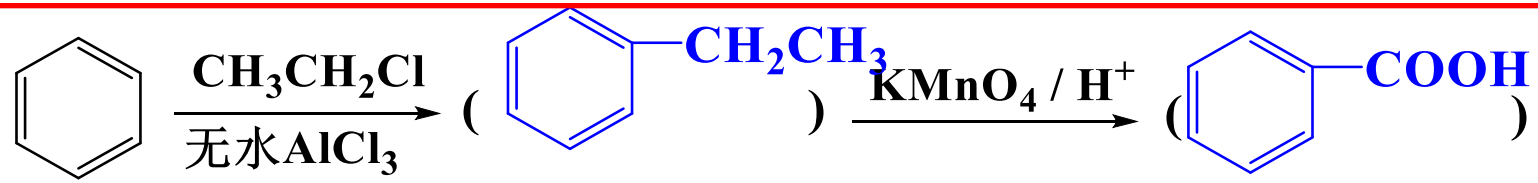
(1).



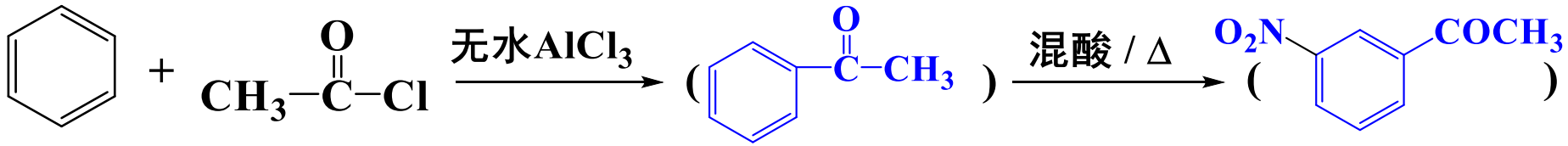
(2).



(3).

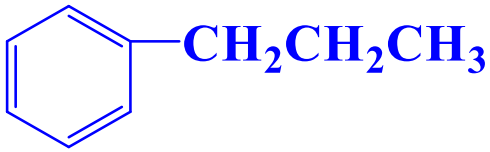


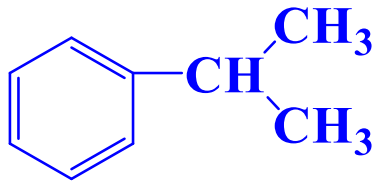
(4).

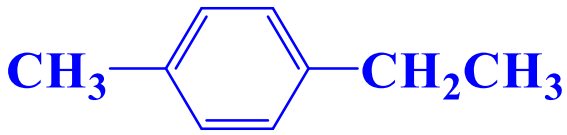


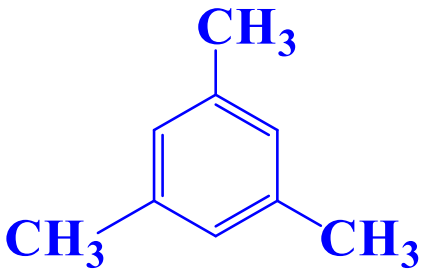
8. 推导结构式

有A、B、C三种芳烃，分子式均为C_9H_{12}，氧化时A生成一元羧酸，B生成二元羧酸，C生成三元羧酸，但硝化时A得到2种主要一元硝基化合物，B得到2种一元硝基化合物，C只得到一种一元硝基化合物，试推断A、B、C的可能结构式.

A. 

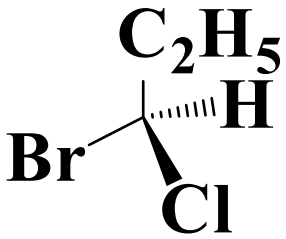
或 

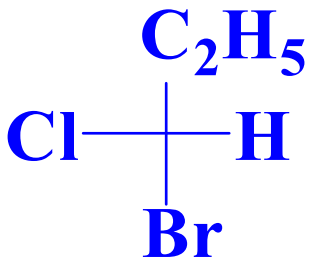
B. 

C. 

1. 写出费歇尔投影式，命名并标记手性碳原子构型

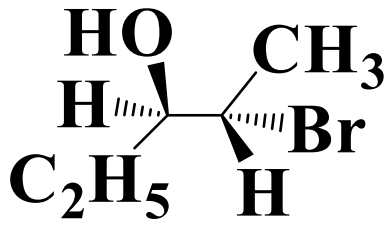
(1).

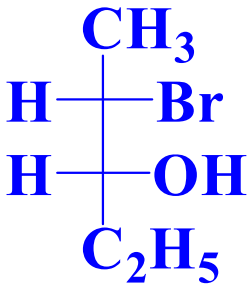




（*S*）-1-氯-1-溴丙烷

(2).

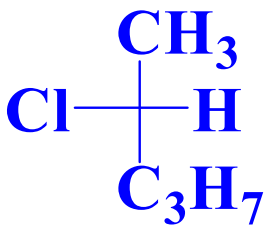




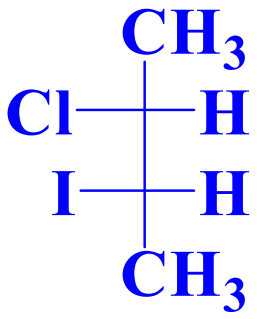
（2*S*，3*R*）-2-溴-3-戊醇

2. 写出下列旋光异构体的费歇尔投影式

(1). (*R*)-2-氯戊烷

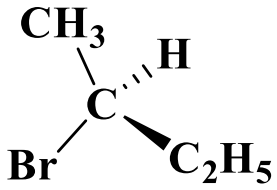


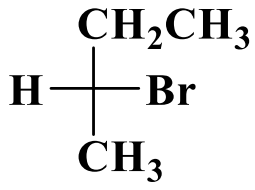
(2). (2*R*, 3*S*)-2-氯−3−碘丁烷

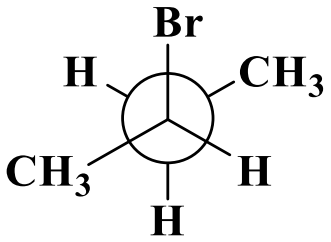


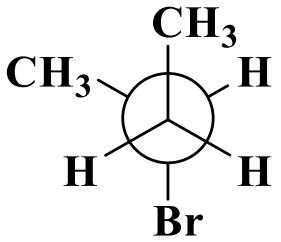
3. 选择

(1). 下列各结构式，与其余结构式不是同一化合物（ B ）

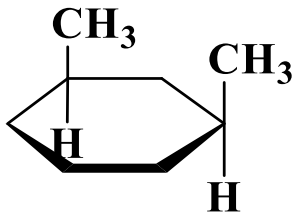
A. 

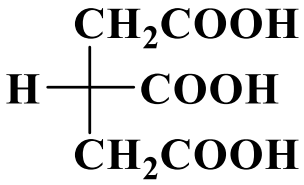
B. 

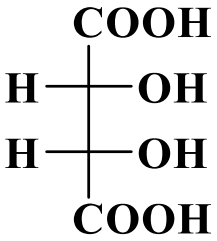
C. 

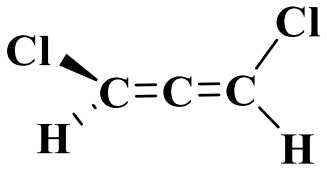
D. 

(2). 下列化合物分子中，有手性的是（ D ）

A. 

B. 

C. 

D. 

(3). 下列哪些化合物既有顺反异构，又有旋光性（ A D ）

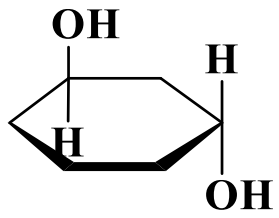
A. $C_6H_5CH=CHCHCOOH$（OH）

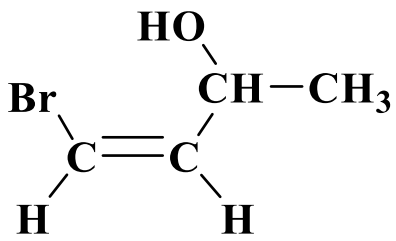
B. $CH_3CH=CHCHCH_3$（CH_3）

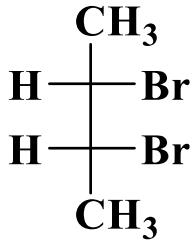
C. $CH_3CH=CHCH_2COOH$

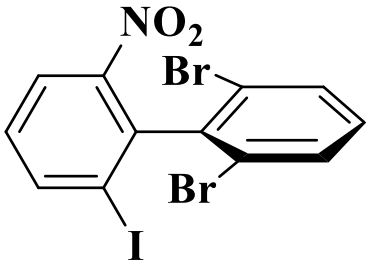
D. $BrCH=CHCHCH_3$（Cl）

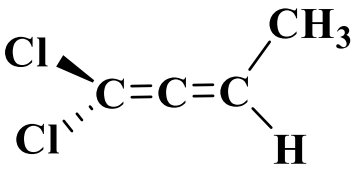
4. 判断下列化合物中哪些具有旋光性

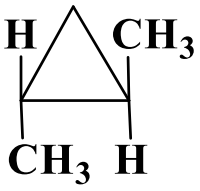
(1) 

(2) 

(3) 

(4) 

(5) 

(6) 

有旋光性：　　(1)　(2)　(6)

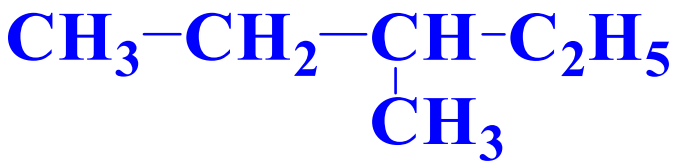
5. 推导结构式

(1). C_6H_{12}是一个有旋光性的不饱和烃，加氢后生成相应的饱和烃。C_6H_{12}不饱和烃是什么？生成的饱和烃有无旋光性？

不饱和烃：



饱和烃：



无旋光性

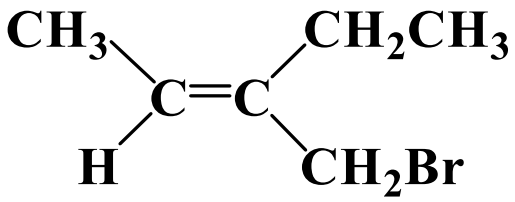
(2). 某化合物(A)的分子式为C_6H_{10}，具有光学活性。可与碱性硝酸银的氨溶液反应生成白色沉淀。若以Pt为催化剂催化氢化，则(A)转变为C_6H_{14}(B)，(B)无光学活性。试推测(A)和(B)的结构式

A: $CH{\equiv}C{-}\underset{\displaystyle CH_3}{\underset{|}{CH}}{-}C_2H_5$

B: $H_3C{-}CH_2{-}\underset{\displaystyle CH_3}{\underset{|}{CH}}{-}C_2H_5$

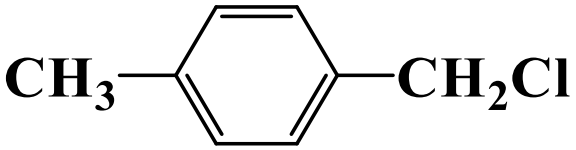
1. 命名或写结构式

(1).



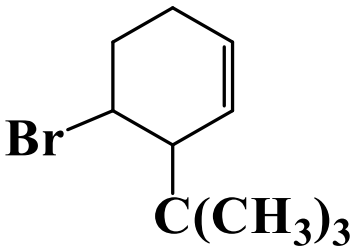
E-2-乙基-1-溴-2-丁烯

(2).



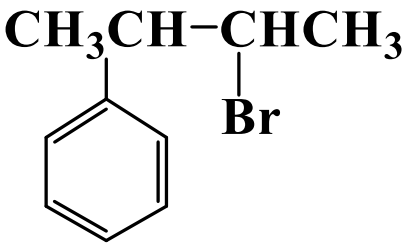
对甲基苯甲基氯

(3).



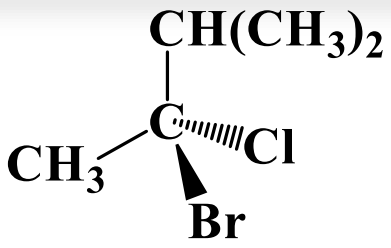
3-叔丁基-4-溴环己烯

(4).



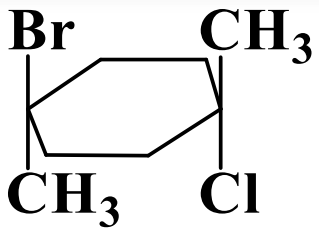
2-苯基-3-溴丁烷

(5).



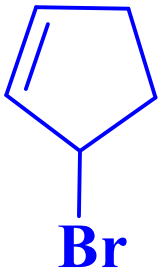
S-2-甲基-3-氯-3-溴丁烷

(6).

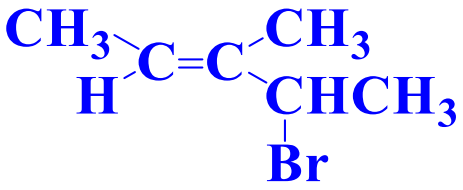


反-1,4-二甲基-1-氯-4-溴环己烷

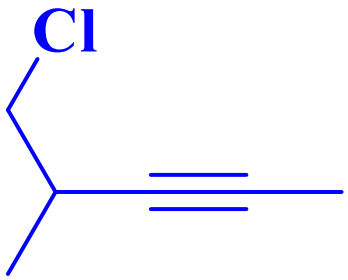
(7). 3-溴环戊烯



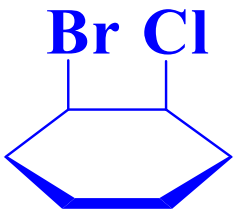
(8). (E) -3-甲基-4-溴-2-戊烯



(9). 4-甲基-5-氯-2-戊炔



(10). 顺-1-氯-2-溴环己烷



2. 完成下列反应式

(1). $(CH_3)_2CH-\underset{Cl}{CH}CH_2CH_3 \xrightarrow[C_2H_5OH,\ \Delta]{KOH} (CH_3)_2C=CHCH_2CH_3$

$\xrightarrow{HCl}$ $(CH_3)_2\underset{Cl}{C}-CH_2CH_2CH_3$ $\xrightarrow{NH_3}$ $(CH_3)_2\underset{NH_2}{C}-CH_2CH_2CH_3$

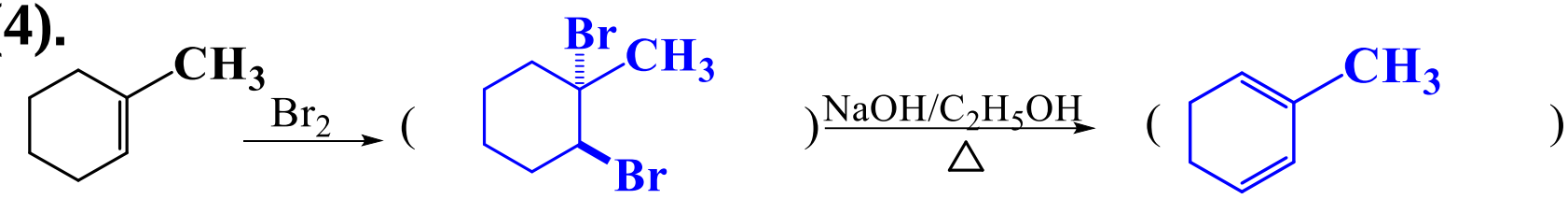
(2). $CH_3CH=CH_2 \xrightarrow{HBr}$ $CH_3\underset{Br}{C}HCH_3$ $\xrightarrow{Mg}$ $CH_3\underset{MgBr}{C}HCH_3$

$\xrightarrow{CO_2}$ $(CH_3)_2CH-\overset{O}{\overset{\|}{C}}-OMgBr$ $\xrightarrow{H_2O}$ $(CH_3)_2CH-\overset{O}{\overset{\|}{C}}-OH$

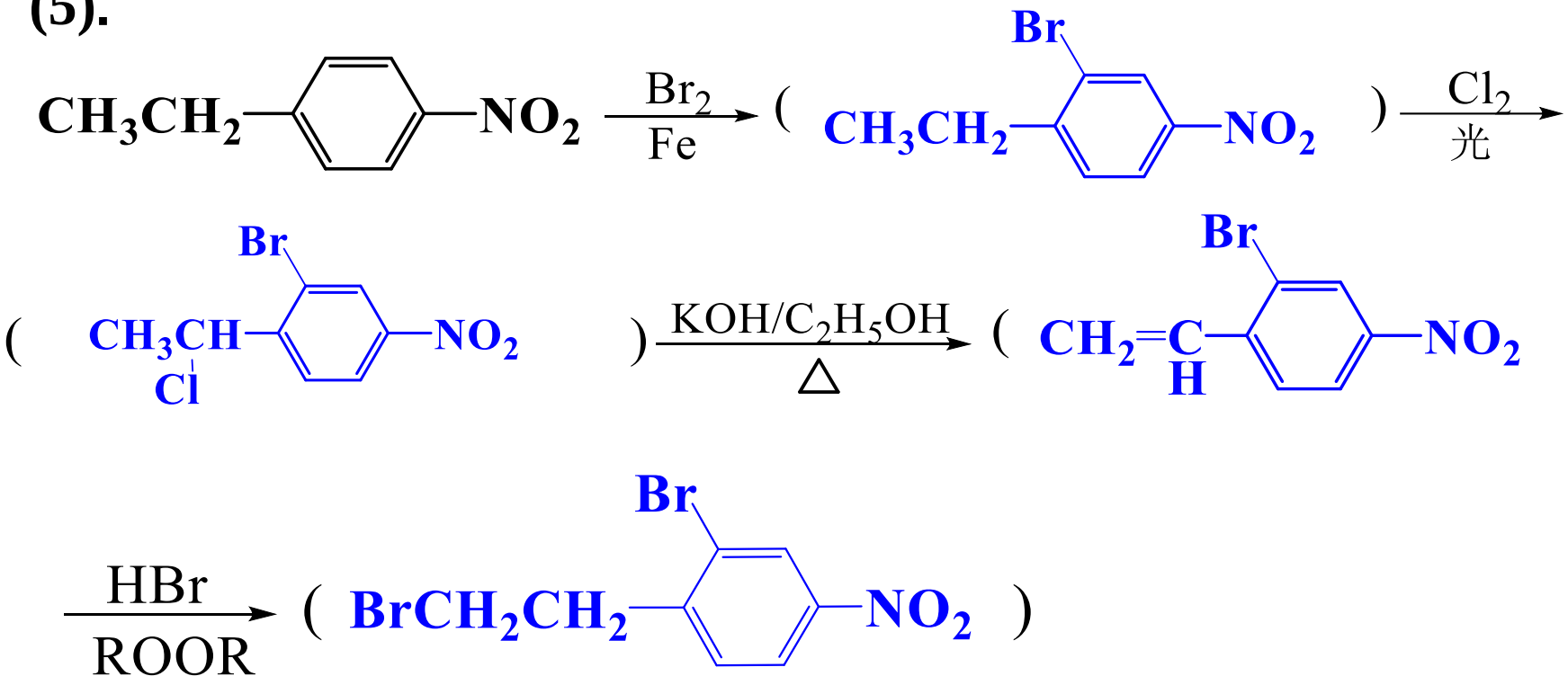
(3).

$ClCH=CHCH_2Cl \xrightarrow[C_2H_5OH]{NaCN} ClCH=CHCH_2CN \xrightarrow{H_3O^+} ClCH=CHCH_2COOH$

(4).


CH3
Br2
Br
CH3
Br
NaOH/C2H5OH
△
CH3

(5).


CH3CH2
NO2
Br2
Fe
Br
CH3CH2
NO2
Cl2
光
Br
CH3CH
Cl
NO2
KOH/C2H5OH
△
Br
CH2=C
H
NO2
HBr
ROOR
Br
BrCH2CH2
NO2

3. 选择或填空

(1). 不饱和卤代烃 ①CH_2=CHX ②CH_2=CH-CH_2X ③CH_2=CH-CH_2-CH_2X中，卤原子活泼性的递增次序是（D.）。

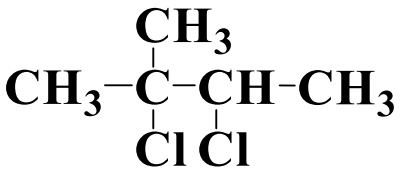
A.①②③ B. ③②① C .②①③ D.①③②

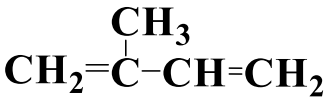
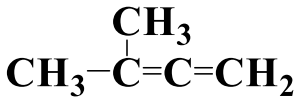
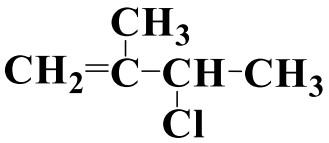
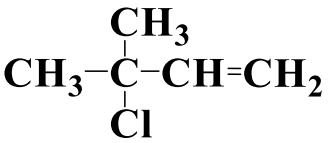
(2). 下列卤代烃中最易发生S_N1反应的是（ C. ）

A. CH_3CH_2Br B. CH_3CH=CHBr C .CH_3CH=$CHCH_2Br$

(3). 在碱性溶液中，按S_N2反应，活性最大的是（ A. ）

A.溴甲烷 B.2-溴丙烷 C.2-溴-2-甲基丁烷 D.1-溴-2-甲基丙烷

(4).  与KOH/乙醇溶液作用的主要产物是（ A. ）

A.  B.  C .  D. 

4. 判断题

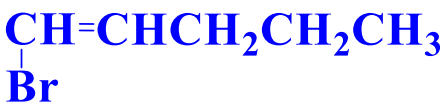
(1). S_N1和S_N2中的“1”与“2”是表示亲核取代反应所经过的步数 （×）

(2). 卤代烃在进行取代反应时，伯卤代烃倾向于发生S_N1反应，而叔卤代烃倾向于发生S_N2反应 （×）

(3). 实验室中可用硝酸银的氨溶液鉴别所有的卤代烃 （×）

5. 用化学方法区别下列各组化合物

(1). 1-溴-1-戊烯　　3-溴-1-戊烯　　4-溴-1-戊烯

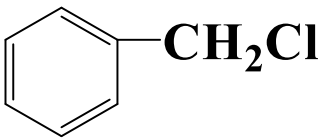


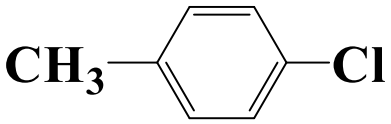


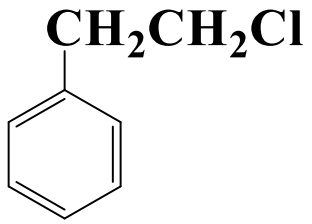


$Ag(NH_3)_2NO_3$ 醇	(-)	↓(淡黄色)	(-)
加热	(-)		↓(淡黄色)

(2). 苄氯　　对氯甲苯　　1-苯基-2-氯乙烷



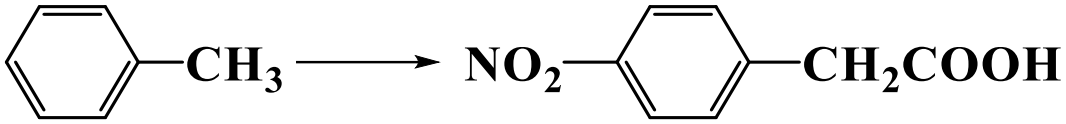


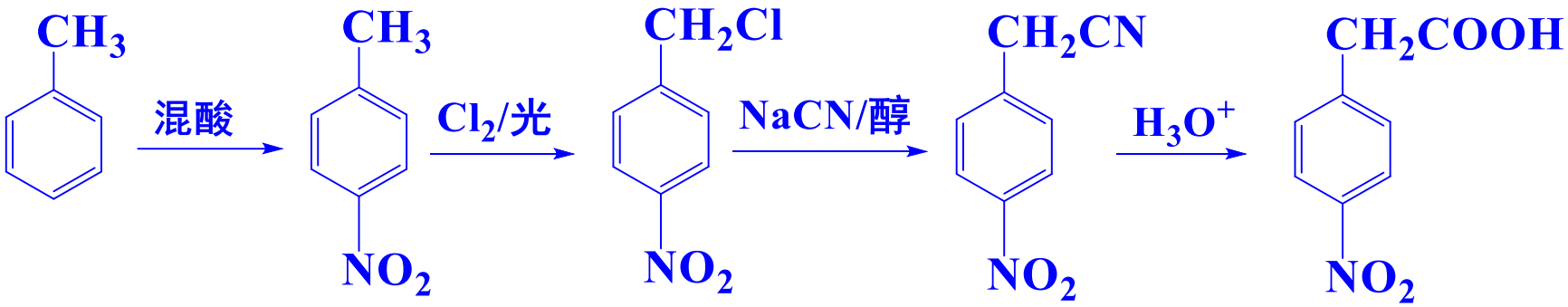


$Ag(NH_3)_2NO_3$ 醇	↓(白色)	(-)	(-)
加热		(-)	↓(白色)

6. 合成下列化合物

(1).

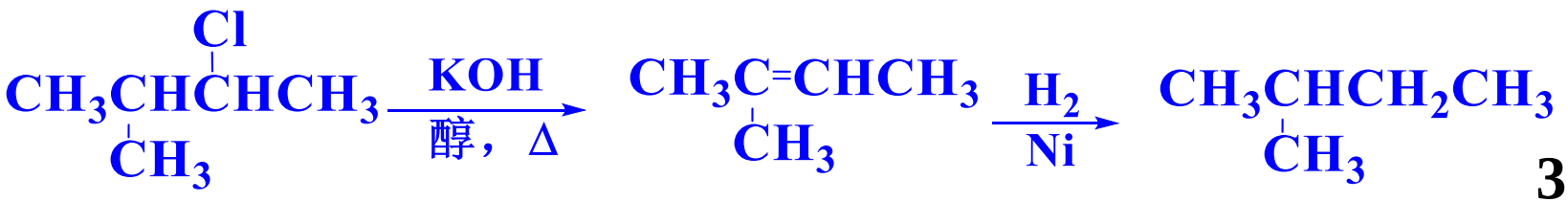




(2). $CH_3CH_2CH{=}CH_2 \longrightarrow CH_3CH{=}CHCH_3$



(3). 2-甲基-3-溴丁烷 ⟶ 2-甲基丁烷



7. 推断题

(1). 某卤代烃A (C_3H_7Br)与KOH乙醇溶液作用生成B (C_3H_6)，B氧化后得到具有两个碳原子的羧酸C和二氧化碳、水，B与溴化氢作用正好得到A 的同分异构体D。试推测A，B，C，D的结构

A. $CH_3CH_2CH_2Br$　　B. $CH_2=CHCH_3$

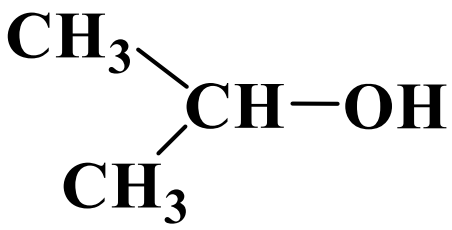
C. CH_3COOH　　D. $CH_3\underset{\underset{Br}{|}}{C}HCH_3$

(2). 某卤代烃分子式为$C_6H_{13}I$，用KOH乙醇溶液处理后将所得到的产物臭氧化，臭氧化产物还原水解生成 $(CH_3)_2CHCHO$和CH_3CHO。试推测该卤代烃的结构。



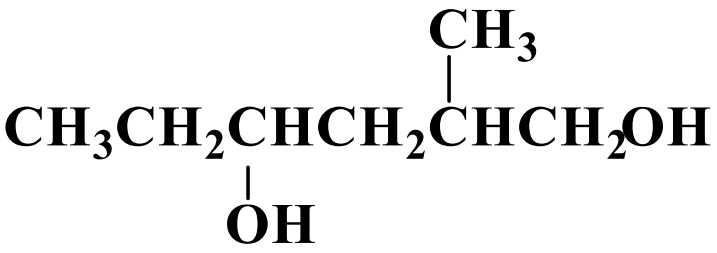
1. 命名或写结构式

(1).



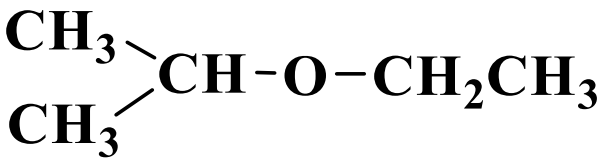
2-丙醇

(2).



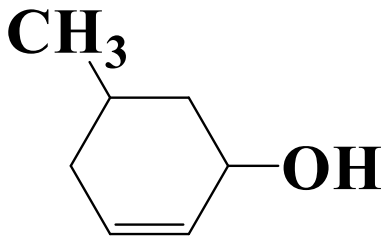
2-甲基-1,4-己二醇

(3).



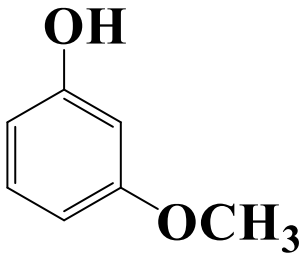
2-乙氧基丙烷
（乙基异丙醚）

(4).



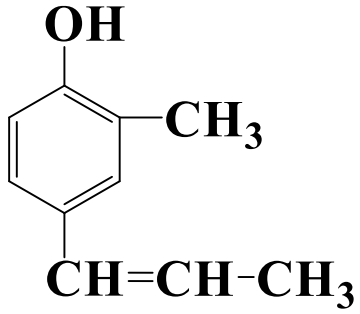
5-甲基-2-环己烯醇

(5).



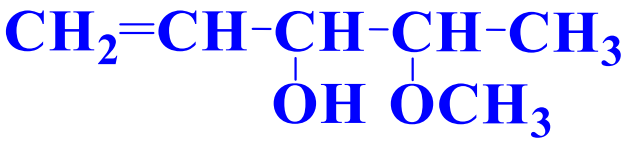
3–甲氧基苯酚

(6).

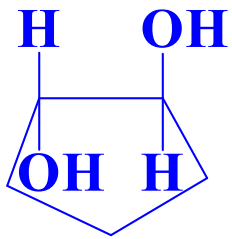


2–甲基–4–丙烯基苯酚

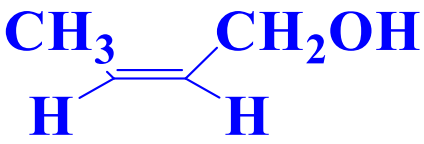
(7). 4–甲氧基–1–戊烯–3–醇



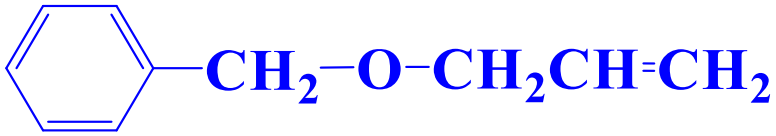
(8). 反–1,2–环戊二醇



(9). Z-2-丁烯-1-醇



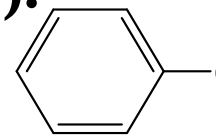
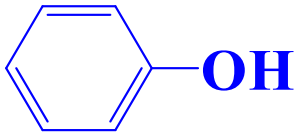
(10). 苄基烯丙基醚



2. 完成下列反应式

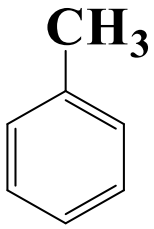
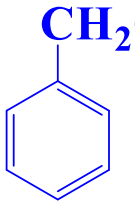
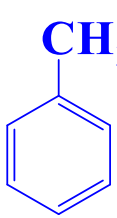
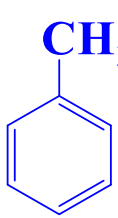
(1). $CH_3CH_2\overset{\overset{CH_3}{|}}{C}(OH)CH_2CH_3 + HBr \longrightarrow (CH_3CH_2\overset{\overset{CH_3}{|}}{C}(Br)CH_2CH_3)$

$\xrightarrow[\text{乙醇}]{NaOH} (CH_3CH_2\overset{\overset{CH_3}{|}}{C}{=}CH_2CH_3)$

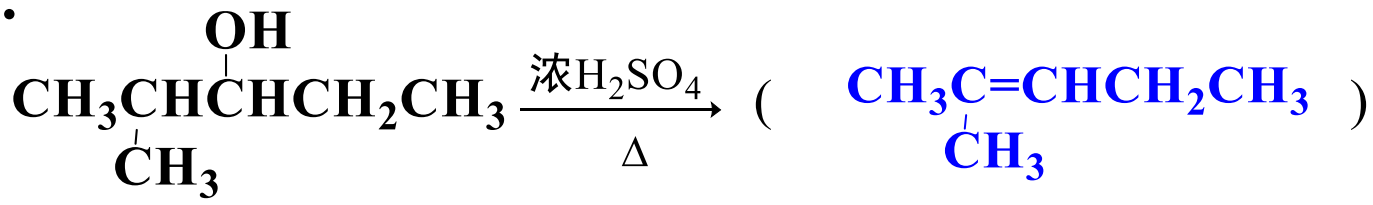
(2).  $+ HI \longrightarrow (CH_3CH_2I) + ($  $)$

(3). $CH_3CH_2\overset{\overset{CH_3}{|}}{C}(OCH_3)CH_3 + 2HI \longrightarrow (CH_3I) + (CH_3CH_2\overset{\overset{CH_3}{|}}{C}(I)CH_3)$

(4).


$\xrightarrow[\text{光}]{Cl_2}$ ()
$\xrightarrow[\text{水}]{NaOH}$ ()
$\xrightarrow[H_2SO_4]{CH_3COOH}$ ()

(5).





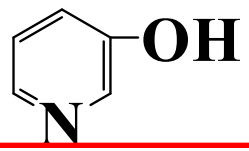
3. 选择或填空

(1). 下列化合物中同时存在π-π共轭、p-π共轭、σ-π超共轭的是（ B ）。

A. $CH_2=CH-CH_2-Cl$

B. $CH_3CH=CH-CH=CH-Cl$

C . $CH_3CH=CH-Cl$

D. 

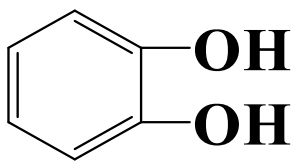
(2). 下列化合物中硝化反应活性最大的是的（ A ），最小的是（ D ）。

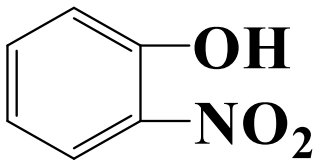
A. 苯酚　　B. 苯　　C. 乙苯　　D. 硝基苯

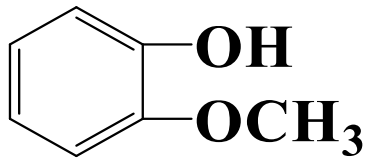
(3). 下列化合物沸点最低的是（ A ）。

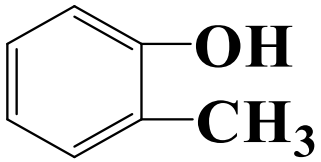
A. 邻氯苯酚　B. 间氯苯酚　C . 对氯苯酚

(4). 下列化合物不能形成分子内氢键的是（ D ）。

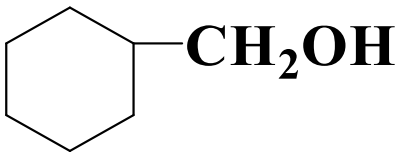
A. 

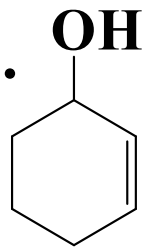
B. 

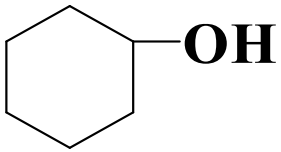
C. 

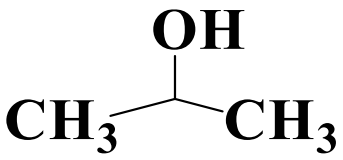
D. 

(5). 下列醇与卢卡斯试剂反应最快的（　　B　　）。

A. 

B. 

C. 

D. 

(6). 醇脱水生成烯烃和醚的反应是消除反应和亲核取代反应的竞争，一般低温时有利于分子间的亲核取代反应生成（　　醚　　），高温时有利于分子内的消除反应生成（　　烯烃　　）。

4. 判断题

(1). 所有醇均易溶于水。（ × ）

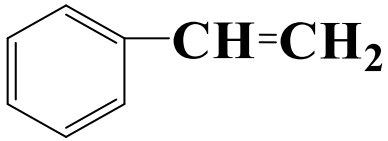
(2). 水的酸性比乙醇弱。（ × ）

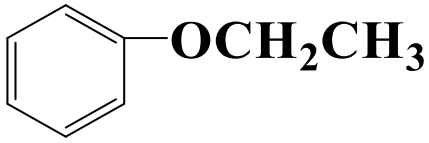
(3). 乙醇和甲醚是官能团异构体。（ √ ）

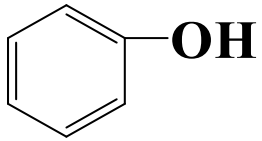
(4). 酚类化合物可以和三氯化铁溶液作用生成有颜色的配合物，常用这种显色反应来鉴别酚类化合物的存在。（ √ ）

5. 用化学方法区别下列各组化合物

(1). 苯乙烯　　苯乙醚　　苯酚





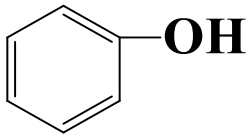


$FeCl_3$　(-)　　(-)　　紫色

Br_2/H_2O　褪色　　(-)

(2). 1-丁醇　　丁醚　　苯酚

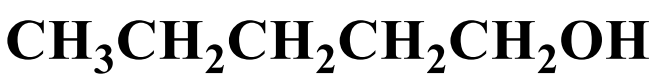
$CH_3CH_2CH_2CH_2OH$　　$C_4H_9OC_4H_9$



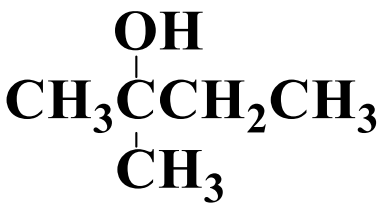
$FeCl_3$　(-)　　(-)　　紫色

Na　$H_2\uparrow$　　(-)

(3). 正戊醇　　3−甲基−2−丁醇　　2−甲基−2−丁醇



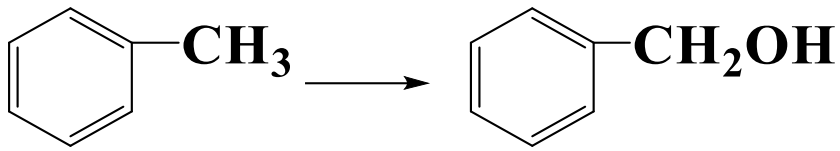


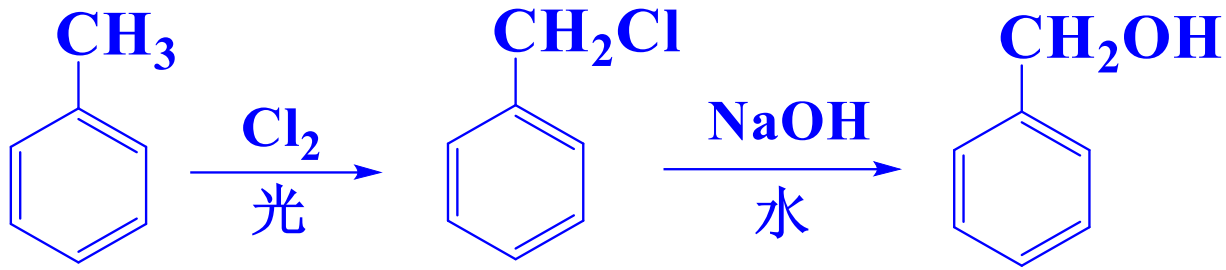


卢卡斯试剂　　加热后混浊　　室温下10分钟后混浊　　室温下1分钟内混浊

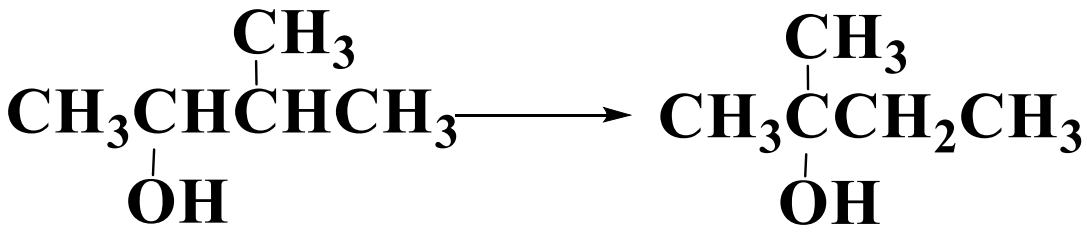
6. 合成下列化合物

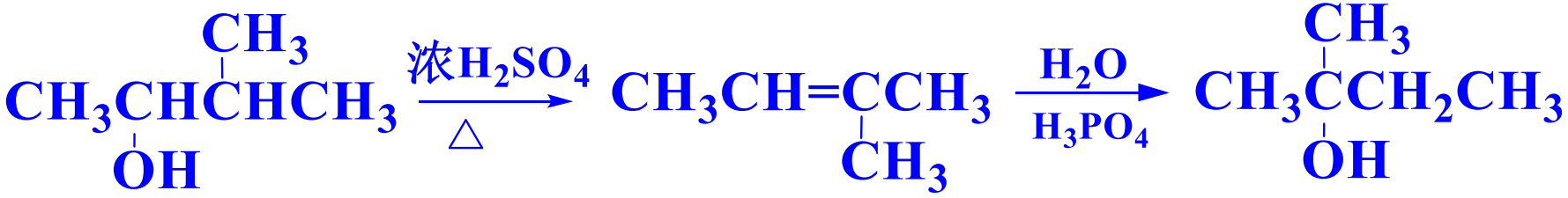
(1).


CH3
CH2OH


CH3
Cl2
光
CH2Cl
NaOH
水
CH2OH

(2).


CH3
CH3CHCHCH3
OH
CH3
CH3CCH2CH3
OH


CH3
CH3CHCHCH3
OH
浓H2SO4
△
CH3CH=CCH3
CH3
H2O
H3PO4
CH3
CH3CCH2CH3
OH

(3). $CH_3CH_2OH \longrightarrow CH_3CH_2CH_2CH_2OH$

$$CH_3CH_2OH \xrightarrow[H^+]{HBr} CH_3CH_2Br \xrightarrow[\text{干醚}]{Mg} CH_3CH_2MgBr$$

$$\xrightarrow[\text{干醚}]{\text{环氧乙烷}} CH_3CH_2CH_2CH_2OMgBr \xrightarrow{H_3O^+} CH_3CH_2CH_2CH_2OH$$

(4). $CH_3CH_2CH_2OH \longrightarrow CH_3CH_2CH_2COOH$

$$CH_3CH_2CH_2OH \xrightarrow[H^+]{HBr} CH_3CH_2CH_2Br \xrightarrow[\text{干醚}]{Mg} CH_3CH_2CH_2MgBr$$

$$\xrightarrow[\text{干醚}]{CO_2} \xrightarrow{H_3O^+} CH_3CH_2CH_2COOH$$

7. 推断题

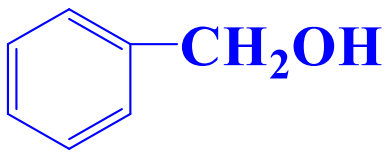
某含氧化合物A，分子式为C_7H_8O，且有两个同分异构体B和C。实验结果如下：

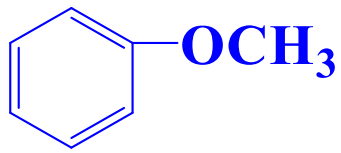
①A、B溶于浓H_2SO_4而不溶于NaOH溶液，C不溶于H_2SO_4但能溶于NaOH，当C全部溶解后加酸酸化，又重新析出；

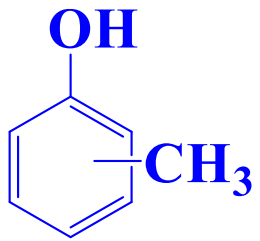
②A与酸性$KMnO_4$溶液作用生成苯甲酸，B无作用，而C则极易被氧化；

③C与$FeCl_3$作用显蓝色，A、B与$FeCl_3$作用均不显色。

试推出A、B、C的结构式。

A. 

B. 

C. 

1. 命名或写结构式

(1). CH_3CHCH_2CHO （C_2H_5 在 CH_3CH 下方）

3–甲基戊醛

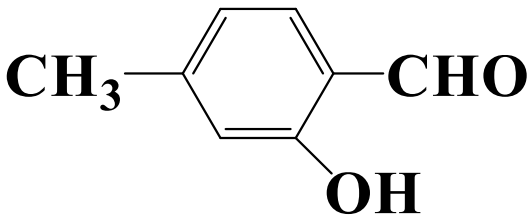
(2). $(CH_3)_2CHCOCH_2CH_3$

2-甲基-3-戊酮

(3). $(CH_3)_2CHCH_2CH{=}CHCHO$

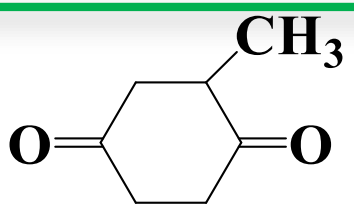
5-甲基-2-己烯醛

(4).



4-甲基-2-羟基苯甲醛

(5).

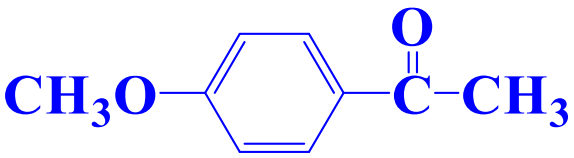


2-甲基-1,4-环己二酮

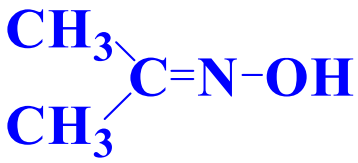
(6). $CH_3COCH{=}CHCOCH_3$

3-己烯-2,5-二酮

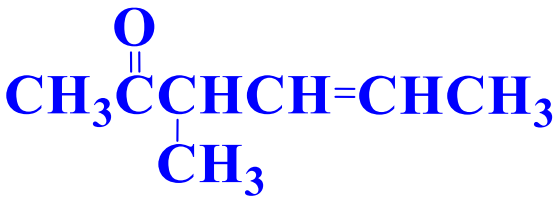
(7). 4-甲氧基苯乙酮



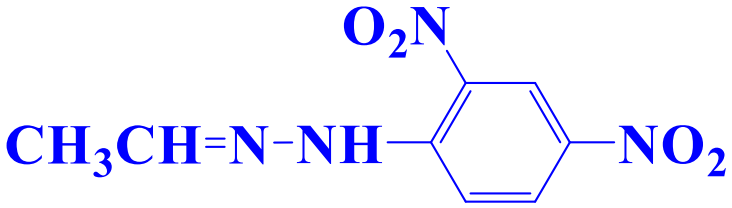
(8). 丙酮肟



(9). 3-甲基-4-己烯-2-酮

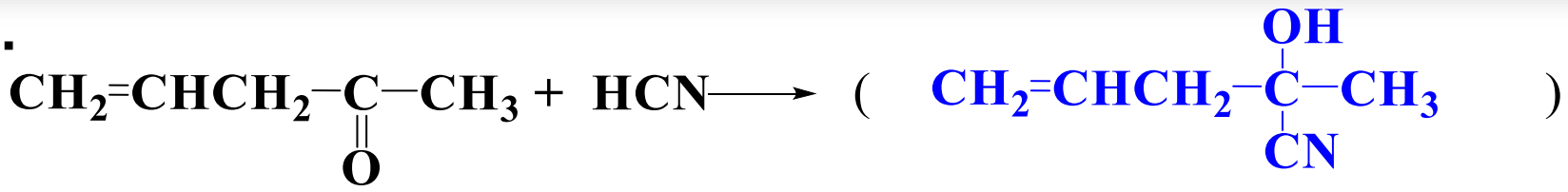


(10). 乙醛-2,4-二硝基苯腙

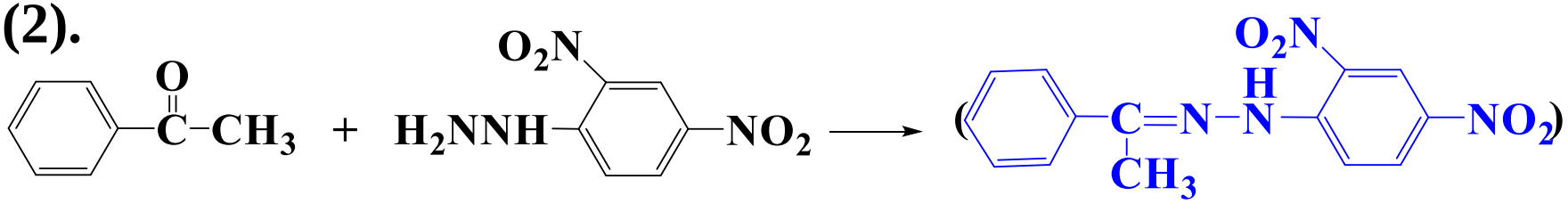


2. 完成下列反应式

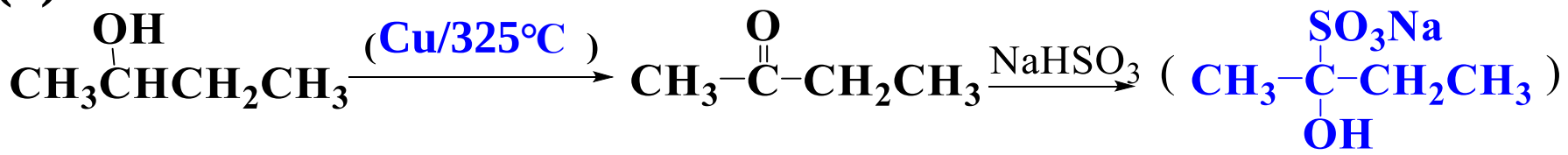
(1).



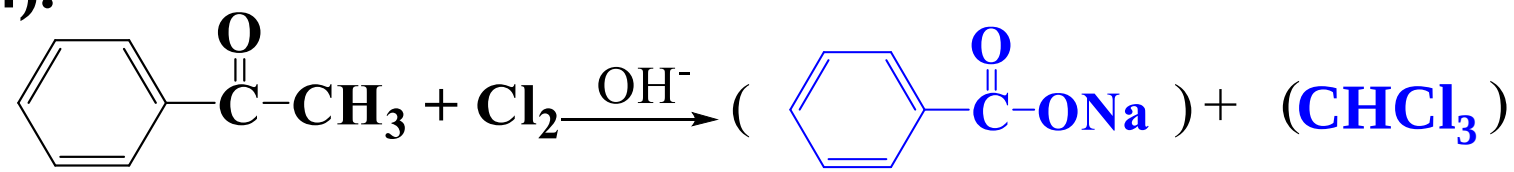
(2).



(3).



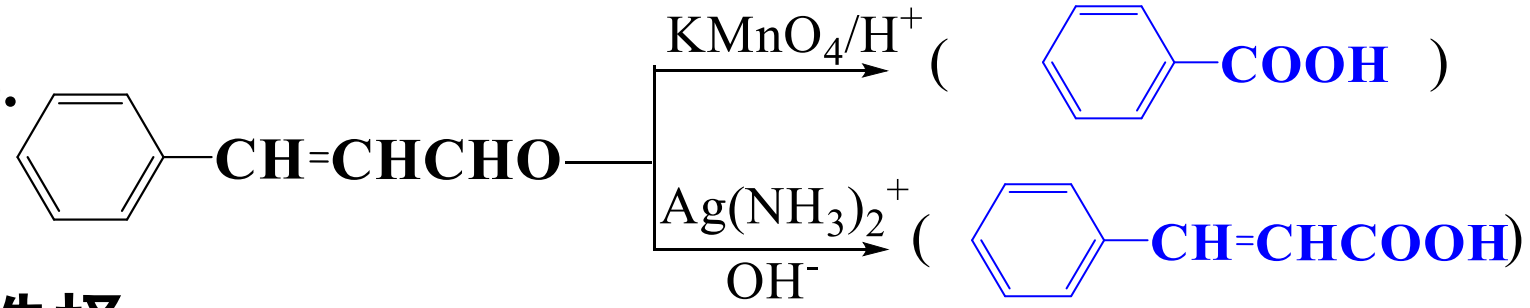
(4).



(5).

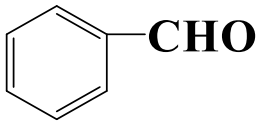
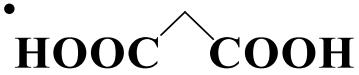
$(CH_3)_3CCHO \xrightarrow{浓NaOH}$ ($(CH_3)_3CCOO^-$) + ($(CH_3)_3CCH_2OH$)

(6).



3. 选择

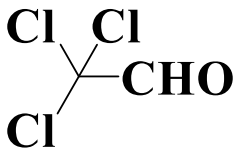
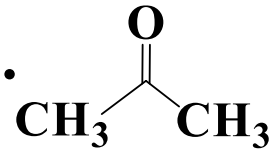
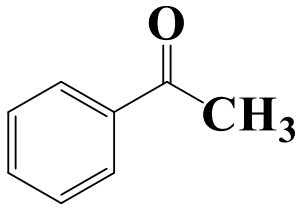
(1). 下列化合物中能发生羟醛缩合反应的是（A.） 能发生歧化反应的是（B.）

A. CH_3-CHO B.  C.  D. 

(2). 为区别乙醛和苯甲醛可选用试剂（ A. . B）；区别乙醛和甲醛可选用试剂（ B. ）；区别乙醇和乙醛可选用试剂（A. C.）

A. 费林试剂 B. I_2/NaOH C . 卢卡斯试剂 D. 格氏试剂

(3). 下列化合物与HCN加成，活性最大的是(A.)；最小的是(D.)。

A.  B. CH_3-CHO C.  D. 

(4). 下列化合物中能发生碘仿反应的是(B.),能与$NaHSO_3$发生加成反应的是(A. D.)。

A．甲醛　　B. 乙醇　　C ． 3-戊酮　　D. 丙醛

(5). 下列化合物与格氏试剂反应生成伯醇的是（ A.）。

A．甲醛　　B. 乙醛　　C ． 丙酮　　D. 苯甲醛

(6). 下列化合物中能与$FeCl_3$起颜色反应的是（ B.）。

A．乳酸　　B. 乙酰乙酸乙酯

C ．$CH_2=CH-CH_2OH$　　D. 苄醇

4. 判断题

(1). 醛和脂肪族甲基酮都可以与HCN和$NaHSO_3$起反应。（ √ ）

(2). 羰基的加成反应属于离子型的亲核加成反应。（ √ ）

(3). 乙醛、丙酮、乙酸、乙酸乙酯都能发生碘仿反应。（×）

(4). 所有醛和酮都可以与$NaHSO_3$发生加成反应。（ × ）

(5). 所有醛和酮都可以与费林试剂反应。（ × ）

(6). 丙酮和异丙醇都可以发生碘仿反应。（ √ ）

5. 用化学方法区别下列各组化合物

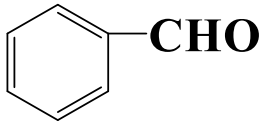
(1). 甲醛　　乙醛　　丙酮

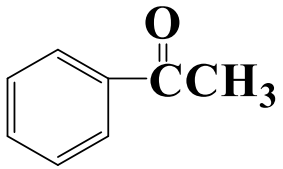
	HCHO	CH_3CHO	$CH_3-\overset{\overset{O}{\|}}{C}-CH_3$
$Ag(NH_3)_2^+$	Ag↓	Ag↓	(-)
$I_2/NaOH$	(-)	CHI_3↓	

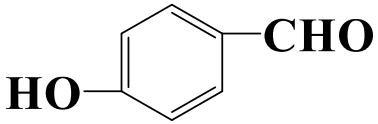
(2). 戊醛　　2−戊酮　　3-戊酮　　2−戊醇

	$CH_3CH_2CH_2CH_2CHO$	$CH_3\overset{\overset{O}{\|}}{C}CH_2CH_2CH_3$	$CH_3CH_2\overset{\overset{O}{\|}}{C}CH_2CH_3$	$CH_3\overset{OH}{\overset{\|}{C}}HCH_2CH_2CH_3$
$Ag(NH_3)_2^+$	Ag↓	(-)	(-)	(-)
$I_2/NaOH$		CHI_3↓	(-)	CHI_3↓
$NaHSO_3$		↓(白色)		(-)

(3). 苯甲醛 苯乙酮 对羟基苯甲醛



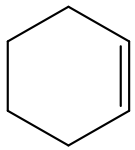


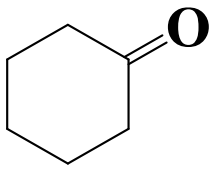


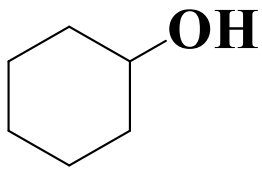
$FeCl_3$ (-) (-) 紫红色

$Ag(NH_3)_2^+$ Ag↓ (-)

(4). 环己烯 环己酮 环己醇







Br_2/H_2O 褪色 (-) (-)

2,4-二硝基苯肼 ↓(黄色) (-)

6. 合成下列化合物

(1). $CH_2=CH_2 \longrightarrow CH_3CH_2CH_2CH_2OH$

$$CH_2=CH_2 \xrightarrow[Ag]{O_2} \triangle_{O}$$

$$CH_2=CH_2 \xrightarrow{HBr} CH_3CH_2Br \xrightarrow[\text{干醚}]{Mg} CH_3CH_2MgBr \xrightarrow[\text{干醚}]{\triangle_{O}} \xrightarrow{H_3O^+} CH_3CH_2CH_2CH_2OH$$

(2). $CH_3CH=CH_2 \longrightarrow CH_3CH_2CH_2CH_2OH$

$$CH_3CH=CH_2 \xrightarrow[\text{过氧化物}]{HBr} CH_3CH_2CH_2Br \xrightarrow[\text{干醚}]{Mg} CH_3CH_2CH_2MgBr$$

$$\xrightarrow[\text{干醚}]{CO_2} \xrightarrow{H_3O^+} CH_3CH_2CH_2COOH \xrightarrow[(2)H_3O^+]{(1)\ LiAlH_4} CH_3CH_2CH_2CH_2OH$$

(2). $CH_3CH_2CHO \longrightarrow CH_3CH_2CHCOOH$ (OH)

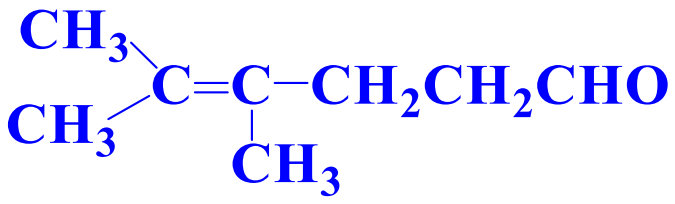
$$CH_3CH_2CHO \xrightarrow{HCN} CH_3CH_2CHCN \xrightarrow{H_3O^+} CH_3CH_2CHCOOH$$
(OH) (OH)

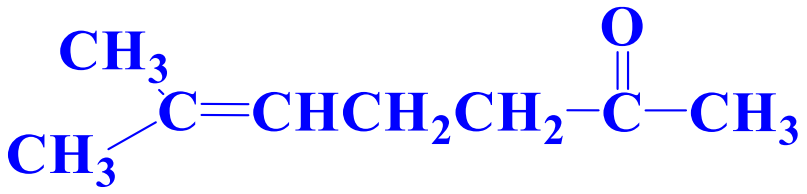
7. 推断题

(1). 某化合物分子式为$C_6H_{12}O$，能与羟胺作用生成肟，但不起银镜反应，在铂的催化下加氢，则得到一种醇，此醇经过脱水、臭氧化、水解等反应后，得到两种液体，其中之一能发生银镜反应，但不起碘仿反应，另一种能起碘仿反应，但不能使Fehling试剂还原，写出该化合物的构造式。

$CH_3CH_2\overset{\overset{O}{\|}}{C}CHCH_3$ (CH_3)

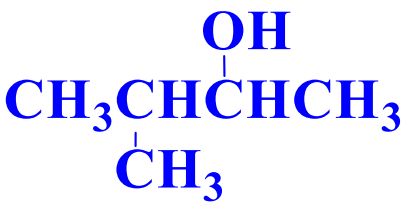
(2). 某化合物A($C_8H_{14}O$)，可以很快的使溴水褪色，也可以和苯肼反应。A氧化后生成一分子丙酮及另一化合物B，B具有酸性，与NaClO (Cl_2+NaOH)反应，生成氯仿及一分子丁二酸。试推测A可能的结构式。





(3). 化合物(A)的分子式为$C_5H_{12}O$，有旋光性，当用碱性$KMnO_4$剧烈氧化时生成没有旋光性的化合物$C_5H_{10}O$(B)。化合物(B)与正丙基溴化镁作用后水解生成旋光性化合物(C)，写出(A)、(B)、(C)的构造式。

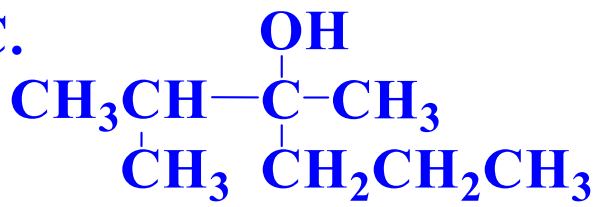
A.



B.



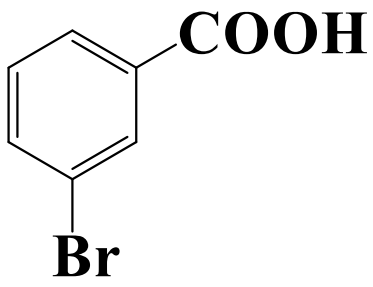
C.



第9章 羧酸、羧酸衍生物和取代酸

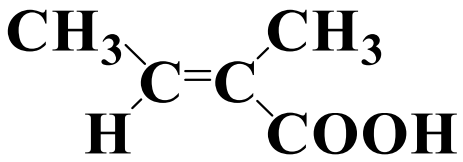
1. 命名或写结构式

(1).



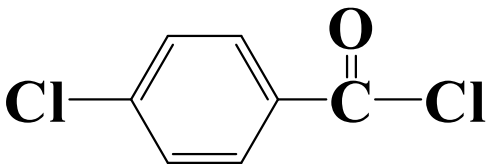
间溴苯甲酸（3-溴苯甲酸）

(2).



顺-2-甲基-2-丁烯酸
（E-2-甲基-2-丁烯酸）

(3).

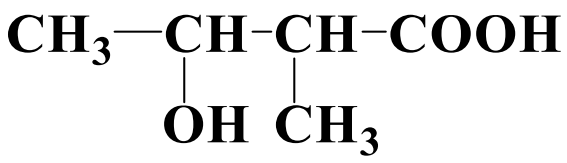


对氯苯甲酰氯

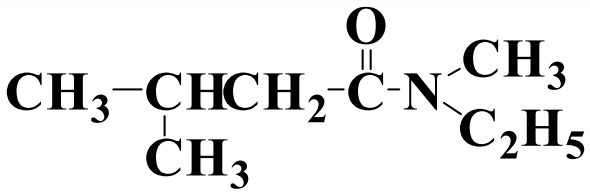
(4).



3-甲基-2-丁烯酸甲酯

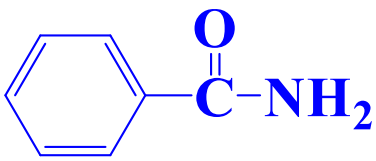
(5). 

2-甲基-3-羟基丁酸

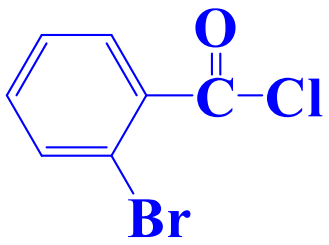
(6). 

N-甲基-N-乙基-3-甲基丁酰胺

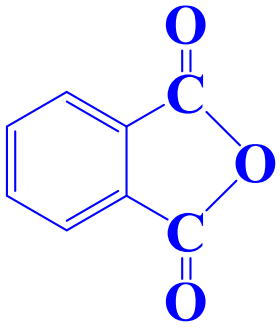
(7). 苯甲酰胺



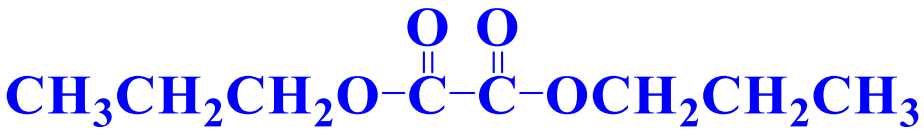
(8). 邻溴苯甲酰氯



(9). 邻苯二甲酸酐



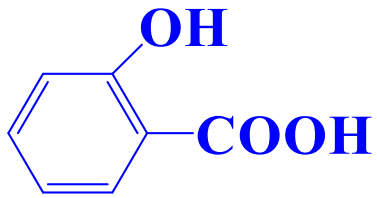
(10). 草酸二丙酯



(11). 乳酸

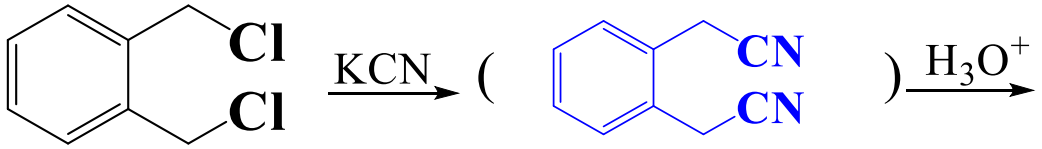


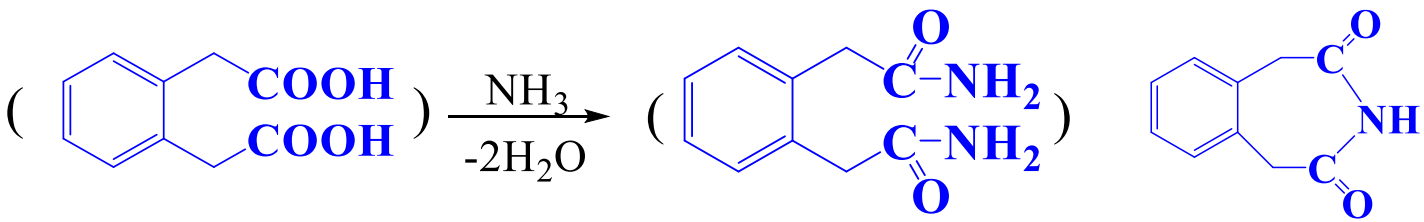
(12). 水杨酸



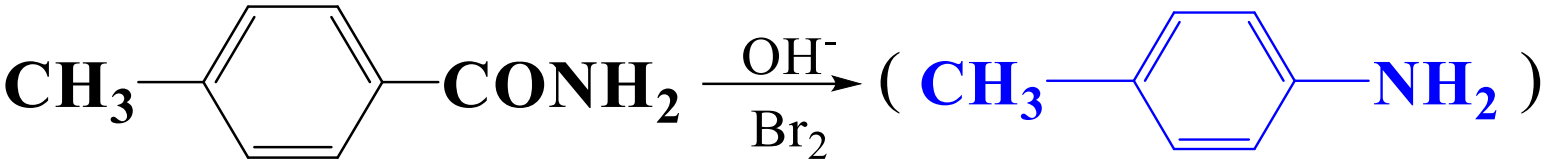
2. 完成下列反应式

(1).

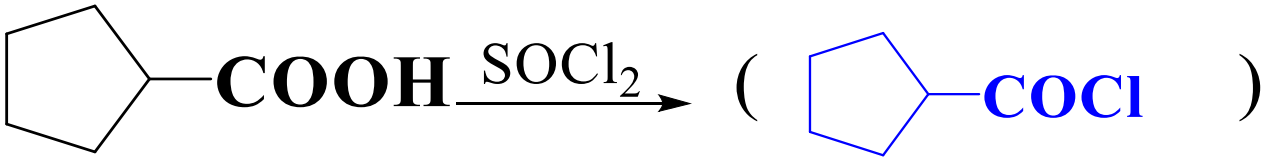




(2).



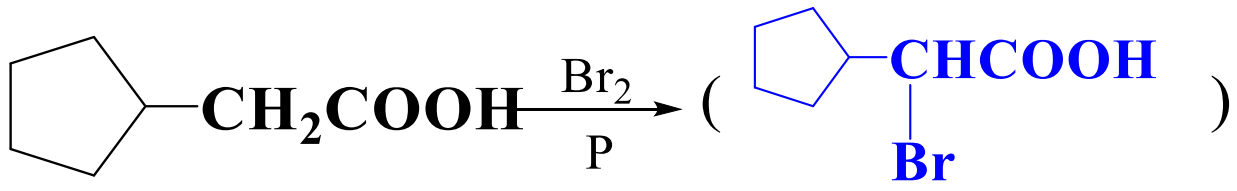
(3).



(4).

$$CH_3COOCH_2CH_3 + (CH_3)_2CHOH \overset{H^+}{\rightleftharpoons} (\ CH_3COOCH(CH_3)_2\ )$$

(5).



(6).



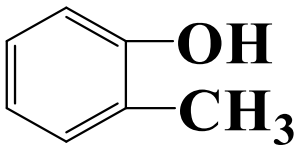
()

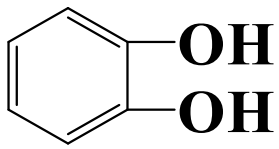
3. 选择或填空

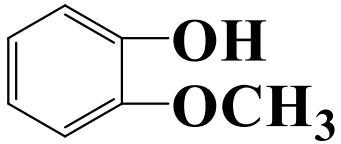
(1). 下列化合物中酸性最强的是（ B ）

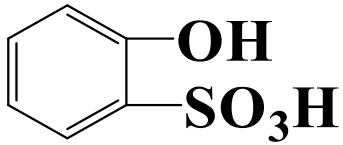
A. 丙酸 B. 2−氯丙酸 C. 3−氯丙酸 D. 2−甲基丙酸

(2). 下列化合物中酸性最强的是（ D ）

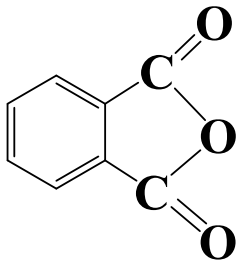
A. 

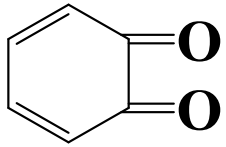
B. 

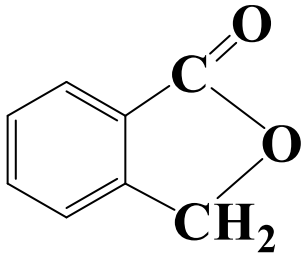
C. 

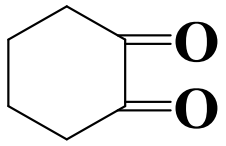
D. 

(3). 下列化合物中（A）属于酸酐，（ C）属于内酯

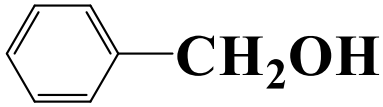
A. 

B. 

C. 

D. 

(4). 下列化合物中能与$FeCl_3$溶液显色的是（ D ）

A. $CH_3CHCOOH$（OH）　B. 

C. $CH_2{=}CHCH_2OH$　D. $CH_3{-}\overset{O}{\overset{\|}{C}}{-}CH_2{-}\overset{O}{\overset{\|}{C}}{-}OCH_2CH_3$

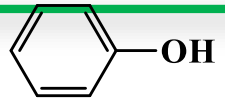
(5). 羧酸衍生物水解反应活性最强的是（ C ）

A. 酯　　B. 酸酐　　C . 酰卤　　D. 酰胺

(6). α-醇酸加热脱水后生成（ 交酯），β−醇酸加热脱水后生成（α，β−不饱和酸），γ−醇酸加热脱水后生成（ 内酯）

(7). 羧酸所连基团的+I效应使酸性（ 减弱 ），-I效应使使酸性（ 增强 ）

4. 判断题

(1).化合物①HCOOH ②CH_3COOH ③$ClCH_2COOH$ ④  ⑤H_2O ⑥CH_3CH_2OH ⑦H_2CO_3酸性递降的次序是③>①>②>⑦>④>⑤>⑥。 （ √ ）

(2). 乙醛、丙酮、乙酸、乙酸酐和乙酰氯都能发生碘仿反应。 （×）

(3). 乙酰氯、乙酸酐、乙酸乙酯、乙酰胺均能发生水解反应。 （ √ ）

(4). 水杨酸与三氯化铁作用不显色。 （ × ）

(5). 酒石酸的系统命名叫2,3−二羟基丁二酸。 （ √ ）

(6). 羧酸的沸点比分子量相近的醇沸点要高。 （ √ ）

(7). 羧酸可被硼氢化钠还原为醇。 （ × ）

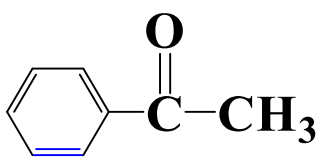
(8). α-羟基酸的酸性比β−羟基酸的酸性强。 （ √ ）

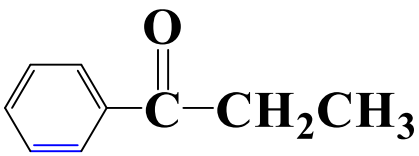
5. 用化学方法区别下列各组化合物

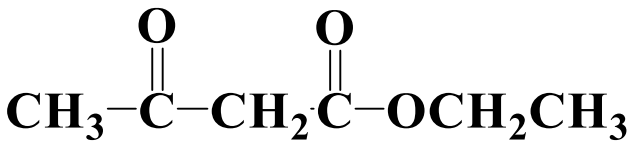
(1). 乙醛　　乙酸　　乙醇　　乙二酸

	CH_3CHO	CH_3CH_2OH	CH_3COOH	$HOOC{-}COOH$
$I_2/NaOH$	$CHI_3\downarrow$	$CHI_3\downarrow$	(-)	(-)
$Ag(NH_3)_2^+$	$Ag\downarrow$	(-)		
加热			(-)	$CO_2\uparrow$

(2).

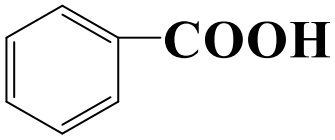


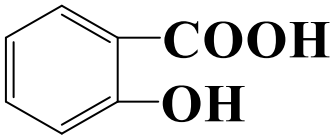


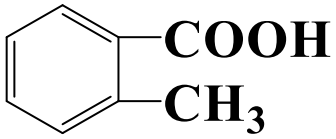


$I_2/NaOH$	$CHI_3\downarrow$	(-)	$CHI_3\downarrow$
$FeCl_3$	(-)		紫色

(3). 苯甲酸　　水杨酸　　邻甲基苯甲酸







$FeCl_3$　　(-)　　紫色　　(-)

$KMnO_4/H^+$　　(-)　　褪色

6. 合成下列化合物

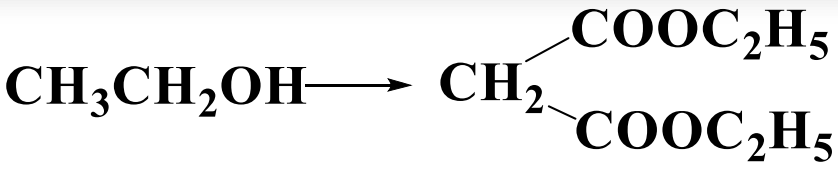
(1). $CH{\equiv}CH \longrightarrow CH_3COOCH_2CH_3$

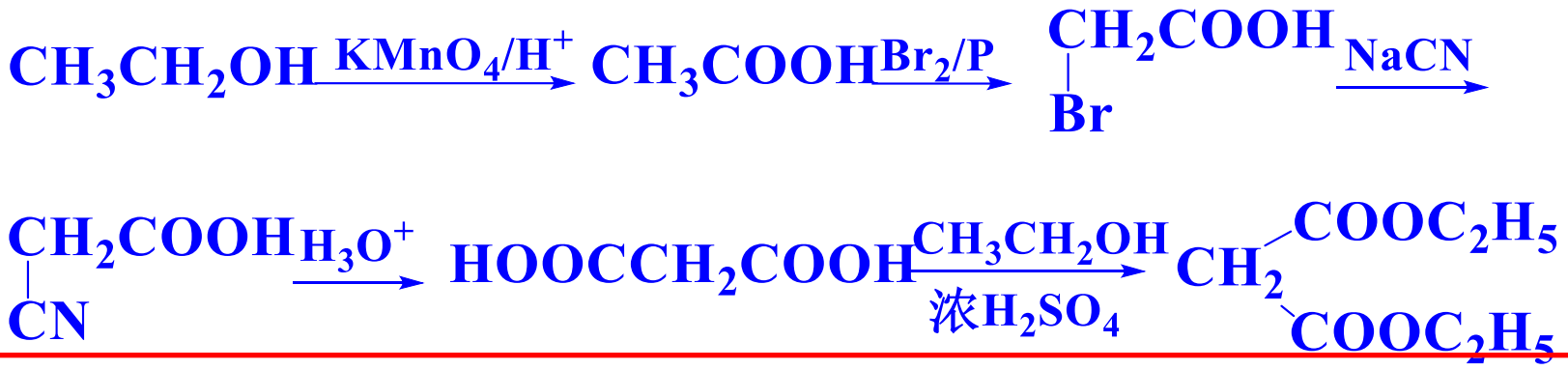
$$CH{\equiv}CH \xrightarrow[H_2O]{Hg^{2+}/H^+} CH_3CHO \xrightarrow[(2)H_3O^+]{(1)NaBH_4} CH_3CH_2OH$$

$$CH_3CHO \xrightarrow{KMnO_4/H^+} CH_3COOH \xrightarrow[H_2SO_4/\Delta]{CH_3CH_2OH} CH_3COOCH_2CH_3$$

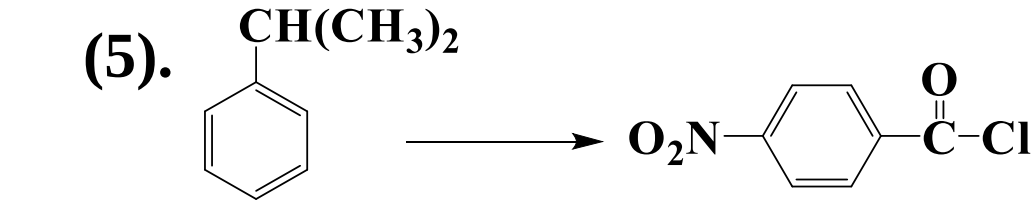
(2). $CH_3CH_2CH{:}CH_2 \longrightarrow CH_3CH_2CHCOCl$
$\quad\quad\quad

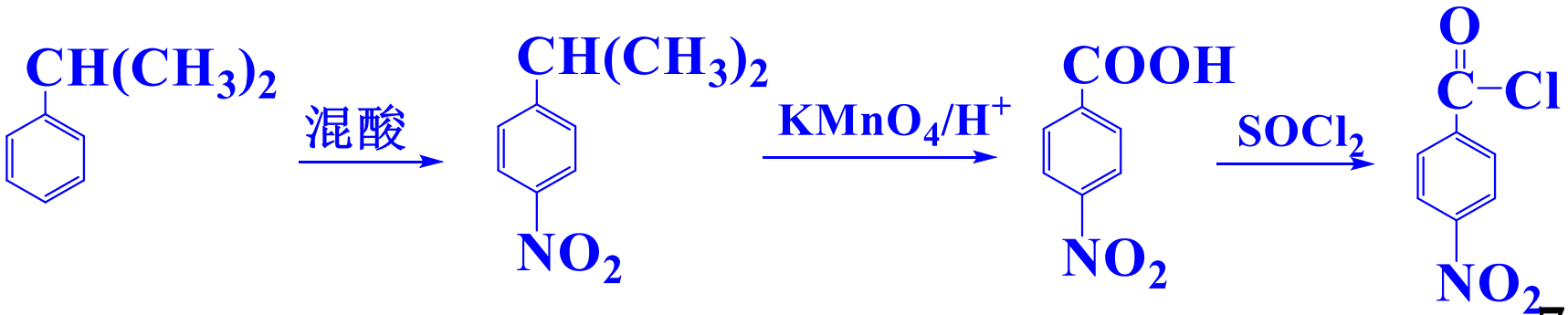
(4).





(5).



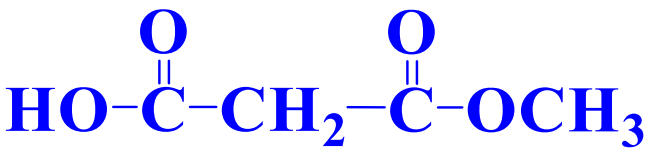


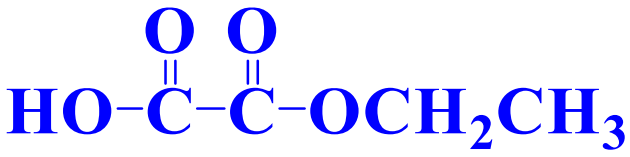
7. 推断题

(1). 化合物A、B的分子式均为$C_4H_6O_4$，它们均可溶于氢氧化钠溶液，与碳酸钠反应放出CO_2，A加热失水成酸酐$C_4H_4O_3$；B加热放出CO_2生成三个碳的酯，试推出A、B的结构式

A. $HOOC\text{-}CH_2\text{-}CH_2\text{-}COOH$

B.





(2). 化合物A、B、C的分子式均为$C_3H_6O_2$，只有A能与$NaHCO_3$作用放出二氧化碳，B和C在氢氧化钠溶液中水解，B的水解产物之一能发生碘仿反应。推测A、B、C的结构式

A: CH_3CH_2COOH

B: $HC(=O)OC_2H_5$

C: $CH_3C(=O)OCH_3$

(3). 有两个酯类化合物A和B，分子式均为$C_4H_6O_2$。A在酸性条件下水解成甲醇和另一个化合物C($C_3H_4O_2$)，C可使Br_2-CCl_4溶液褪色。B在酸性条件下水解生成一分子乙酸和化合物D；D可以发生碘仿反应，也可与土伦试剂作用。试推断A~D的结构

A: $CH_2=CH-\overset{\overset{O}{\|}}{C}-OCH_3$

B: $CH_3\overset{\overset{O}{\|}}{C}-O-CH=CH_2$

C: $CH_2=CH-\overset{\overset{O}{\|}}{C}-OH$

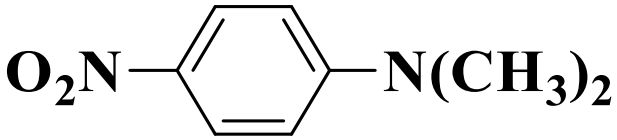
D: $CH_3\overset{\overset{O}{\|}}{C}-H$

第10章　含氮和含磷有机化合物

1. 命名或写结构式

(1). $(CH_3)_2CHNH_2$

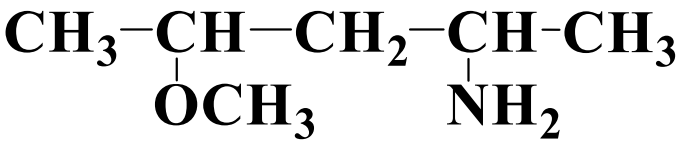
异丙胺

(2). 

N,N−二甲基对硝基苯胺

(3). $(C_2H_5)_2\overset{+}{N}HCH_3OH^-$

氢氧化甲基二乙基铵

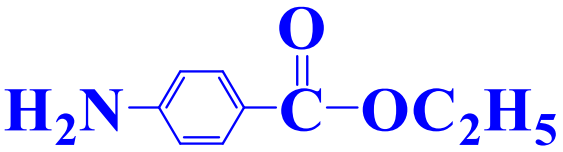
(4). 

2−氨基−4−甲氧基戊烷

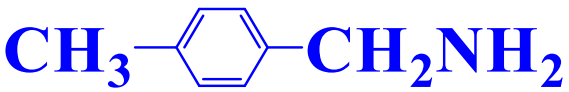
(5). 1,6-己二胺



(6). 对氨基苯甲酸乙酯



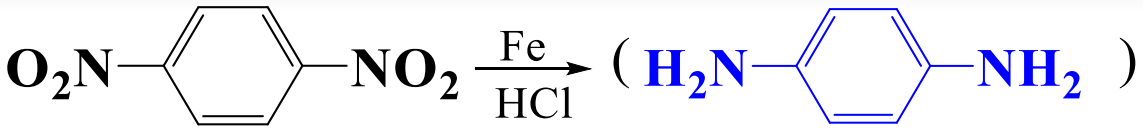
(7). 对甲基苄胺

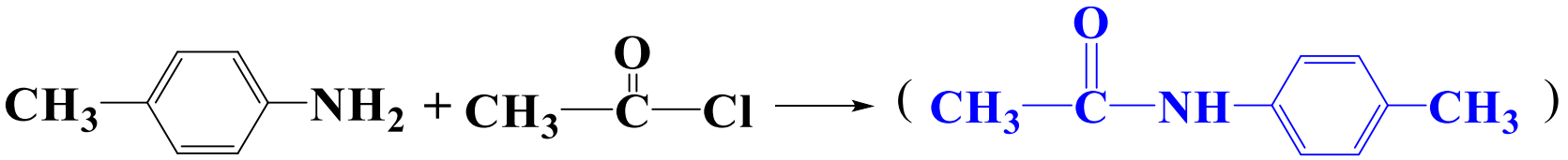


(8). 碘化三甲基羟乙基铵

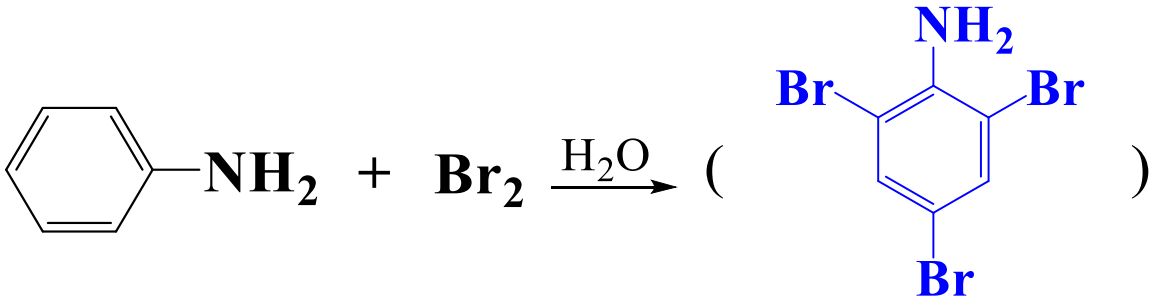


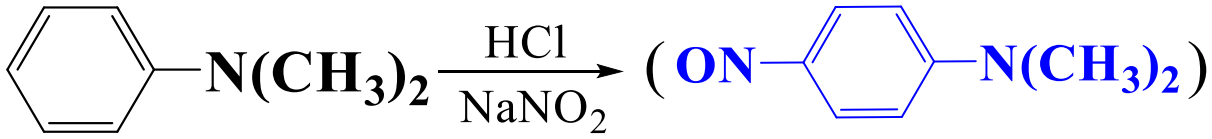
2. 完成下列反应式

(1). 

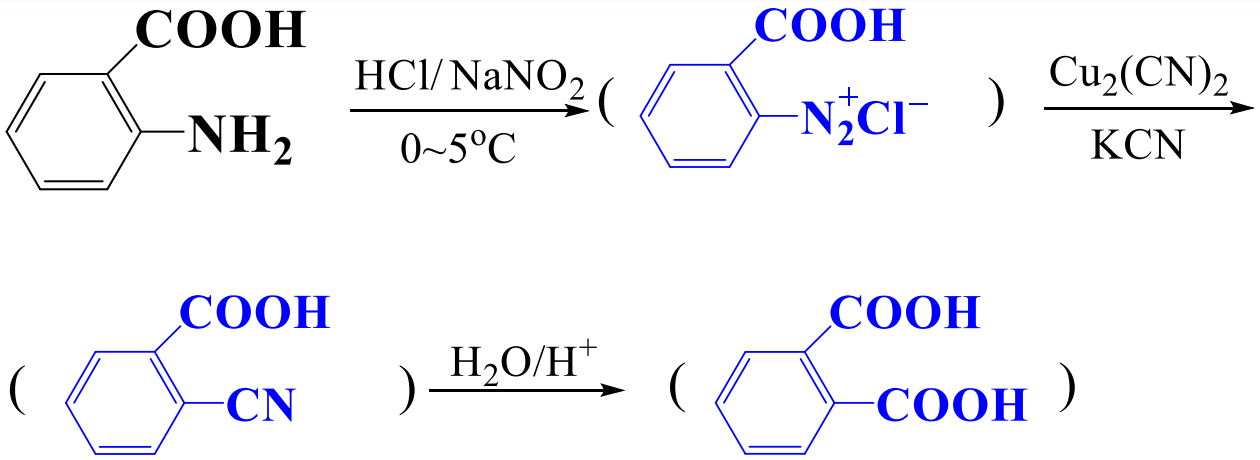
(2). 

(3). $(C_2H_5)_3N + CH_3-\underset{\underset{Br}{|}}{CH}-CH_3 \longrightarrow ((C_2H_5)_3N^+CH(CH_3)_2Br^-)$

(4). 

(5). 

(6).



3. 选择或填空

(1). 下列化合物① CH_3CONH_2 ② $PhNH_2$ ③ $(CH_3CH_2)_2NH$ ④ NH_3 ⑤$CH_3CH_2NH_2$ ⑥ Ph_2NH ⑦ $(CH_3CH_2)_4N^+OH^-$按碱性由强到弱的顺序排列（ (7)＞(3)＞(5)＞(4)＞(2)＞(6)＞(1) ）

(2). 氯化重氮苯属于（ C ）

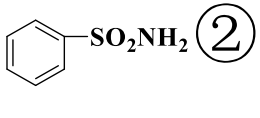
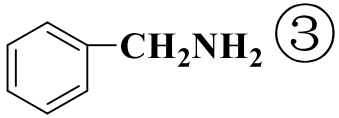
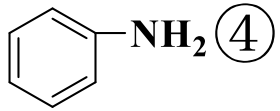
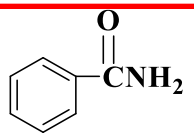
A. 胺　　B. 铵　　C . 重氮盐　　D. 偶氮化合物

(3). 下列化合物中碱性最弱的是（ D），碱性最强的是（ B ）

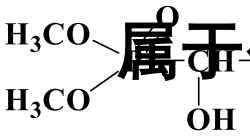
A. 氨　B. 二乙胺　C . 苯胺　D. 邻苯亚甲酰亚胺

4. 判断题

(1). 胺在水溶液中，相对碱性为：叔胺>仲胺>伯胺。　（ × ）

(2). 化合物①  ②  ③  ④  碱性递降的次序为②>③>④>①。　（ √ ）

(3). 凡含有酰胺键  的化合物均能发生缩二脲反应 （ × ）

(4). 敌百虫  属于磷酸酯类农药。　（ × ）

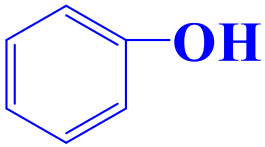
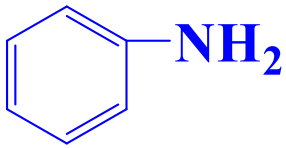
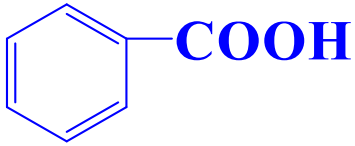
(5). 叔丁胺分子中氨基与三级碳原子直接相连，所以叔丁按属于三级胺。（　× ）

5. 用化学方法区别下列各组化合物

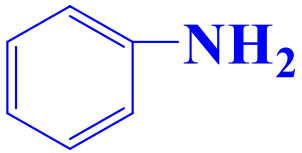
(1). 甲胺　　二甲胺　　三甲胺

	CH_3NH_2	$(CH_3)_2NH$	$(CH_3)_3N$
$HCl/NaNO_2$	$N_2\uparrow$	淡黄色油状物	溶液成一相(无分层)

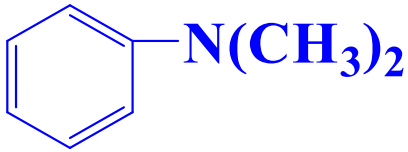
(2). 苯胺　　苯酚　　苯甲酸

			
$FeCl_3$	显紫色	(-)	(-)
Br_2/H_2O		白色↓	(-)

(3). 苯胺　　N-甲基苯胺　　N,N-二甲基苯胺



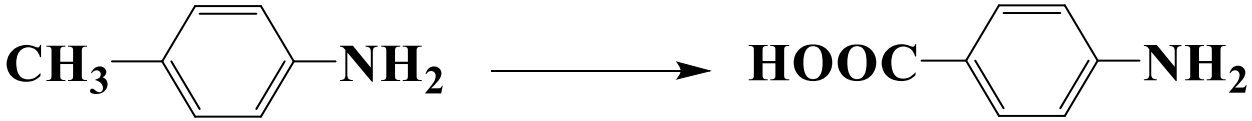


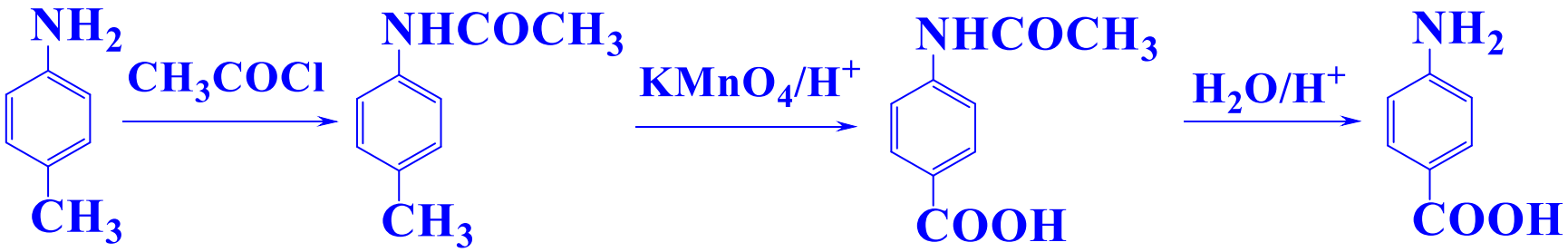


$HCl/NaNO_2$　　$N_2\uparrow$　　淡黄色油状物　　绿黄色片状晶体

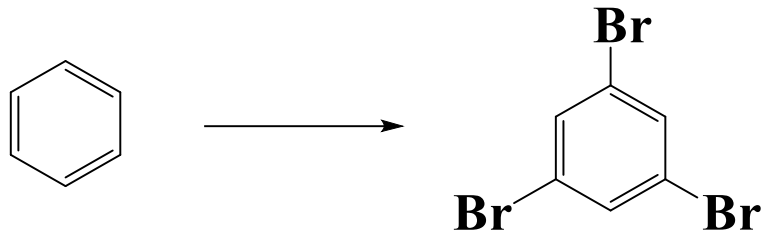
6. 合成下列化合物

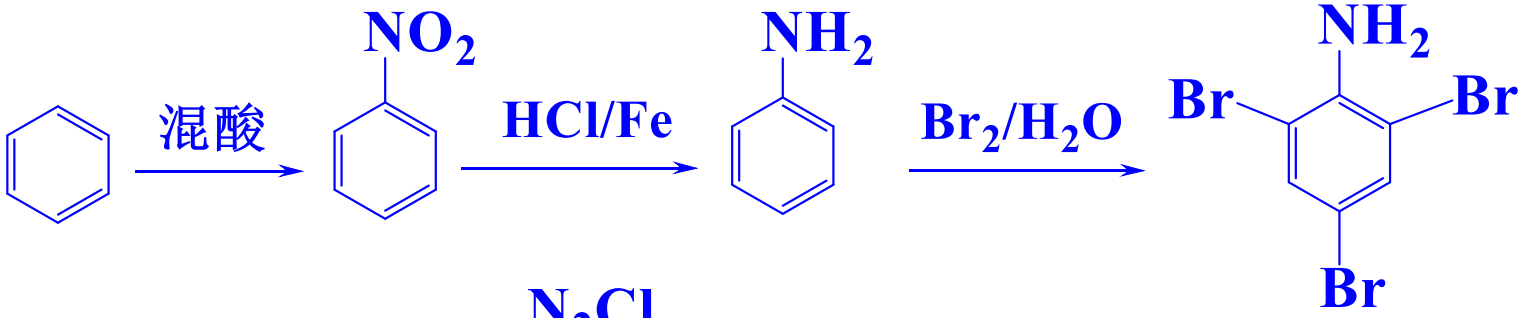
(1).

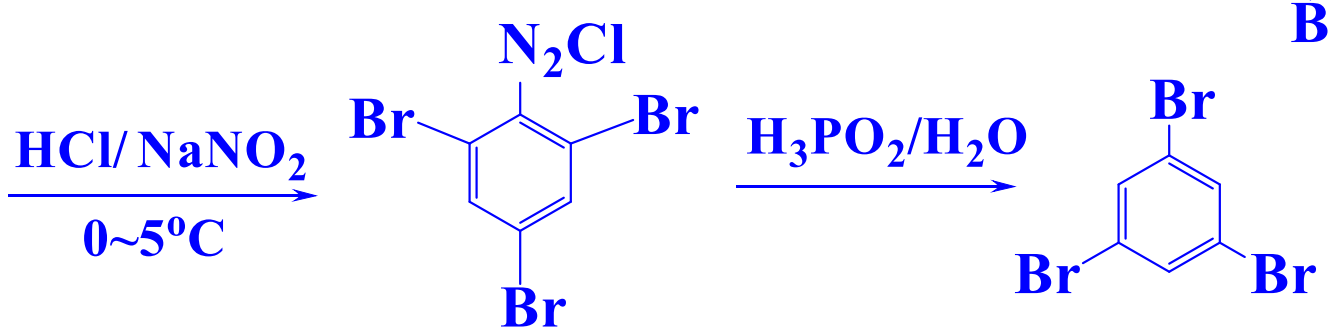




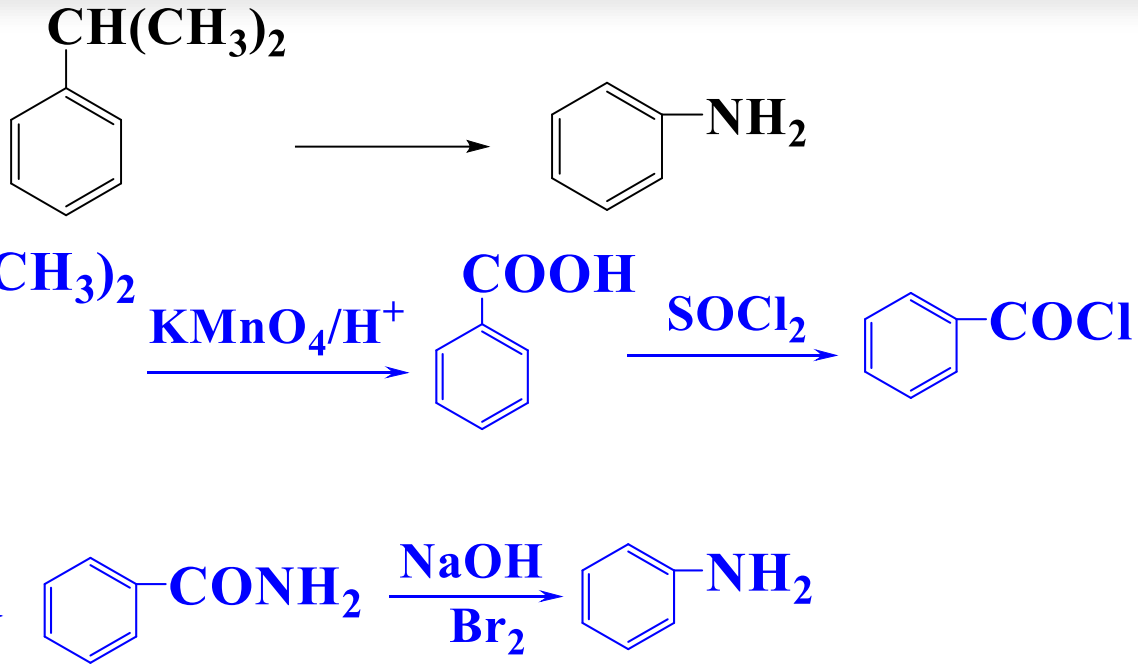
(2).


Br
Br
Br

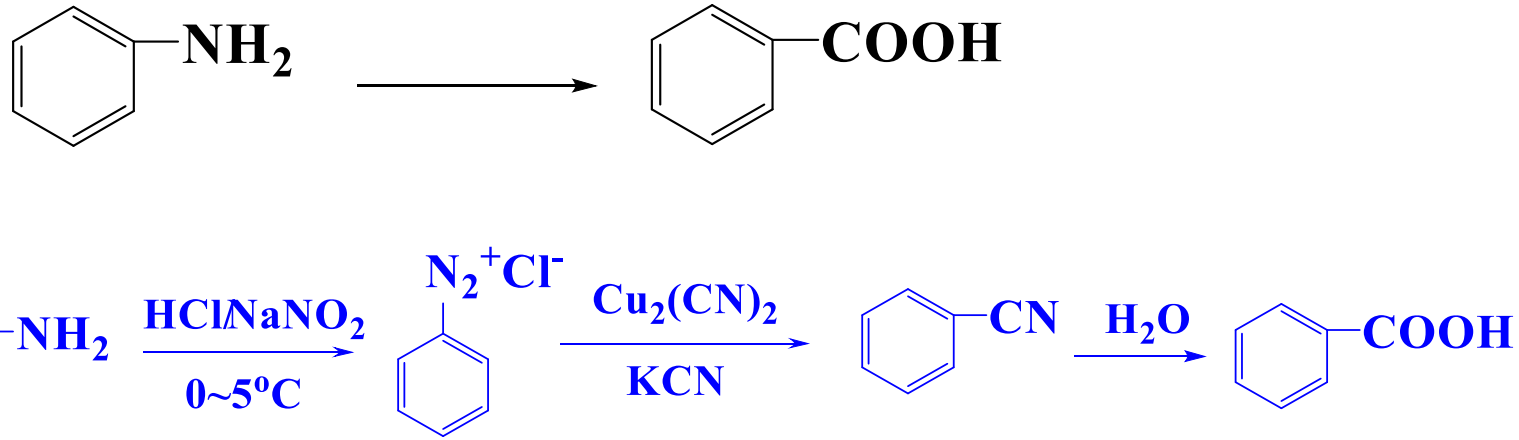

NO2
混酸
HCl/Fe
NH2
Br2/H2O
NH2
Br
Br
Br


HCl/ NaNO2
0~5°C
N2Cl
Br
Br
Br
H3PO2/H2O
Br
Br
Br

(3).


CH(CH3)2
NH2
CH(CH3)2
KMnO4/H+
COOH
SOCl2
COCl
NH3
CONH2
NaOH
Br2
NH2

(4).


NH2
COOH
NH2
HCl/NaNO2
0~5°C
N2+Cl-
Cu2(CN)2
KCN
CN
H2O
COOH

7. 推断题

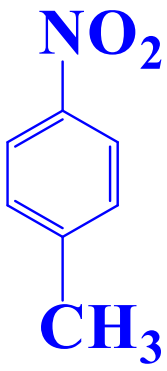
(1). 化合物A的分子式为$C_6H_{15}N$，能溶于稀盐酸，与亚硝酸在室温时作用放出氮气并得到化合物B，B能进行碘仿反应，B与浓硫酸共热得到C(C_6H_{12})，C能使高锰酸钾溶液退色，并氧化分解为乙酸和2-甲基丙酸。试推定A、B、C的构造式，并写出有关反应式。

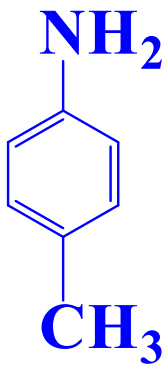
A: $CH_3\underset{\displaystyle NH_2}{\underset{|}{C}}HCH_2CH(CH_3)_2$

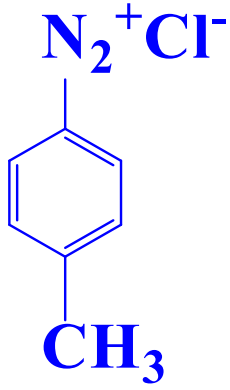
B: $CH_3\underset{\displaystyle OH}{\underset{|}{C}}HCH_2CH(CH_3)_2$

C: $CH_3CH{=}CHCH(CH_3)_2$

(2). 某化合物A分子式为$C_7H_7NO_2$，无碱性。A加氢还原后得到分子式为C_7H_9N的化合物B，B显碱性。在低温和强酸存在下，B可与HNO_2作用生成分子式为$C_7H_7N_2Cl$的化合物C，C加热后可放出N_2并生成对甲苯酚，试推出A、B、C的结构式

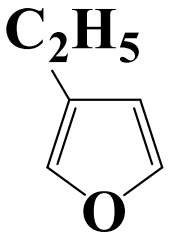
A: 

B: 

C: 

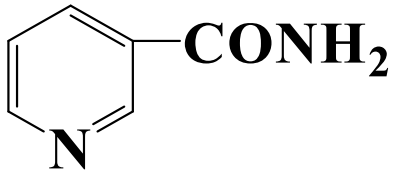
1. 命名或写结构式

(1).



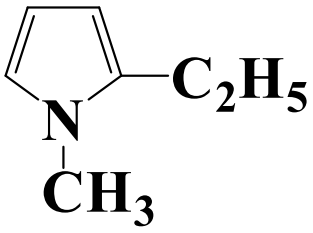
3–乙基呋喃

(2).



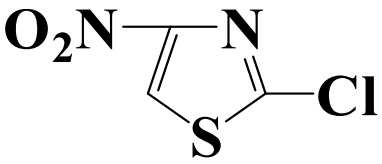
3–吡啶甲酰胺

(3).



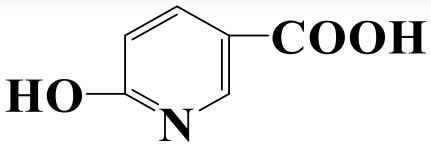
1–甲基–2–乙基吡咯

(4).



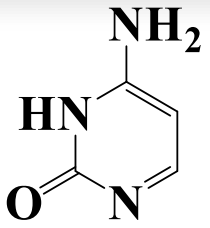
4–硝基–2氯噻唑

(5).



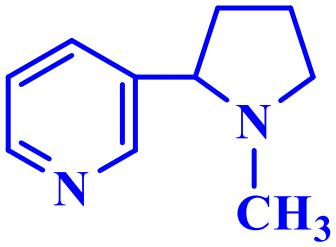
6-羟基-3-吡啶甲酸

(6).

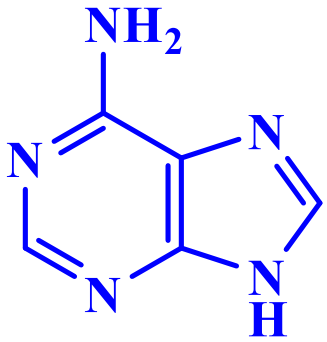


胞嘧啶

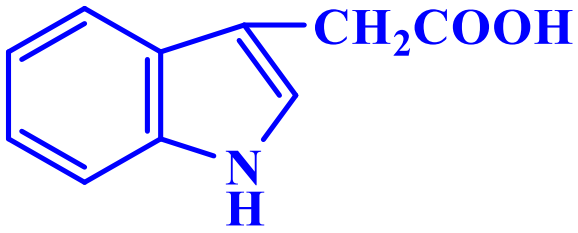
(7). 烟碱



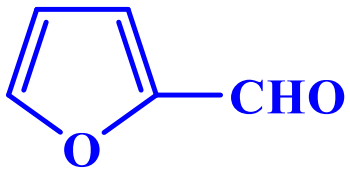
(8). 腺嘌呤



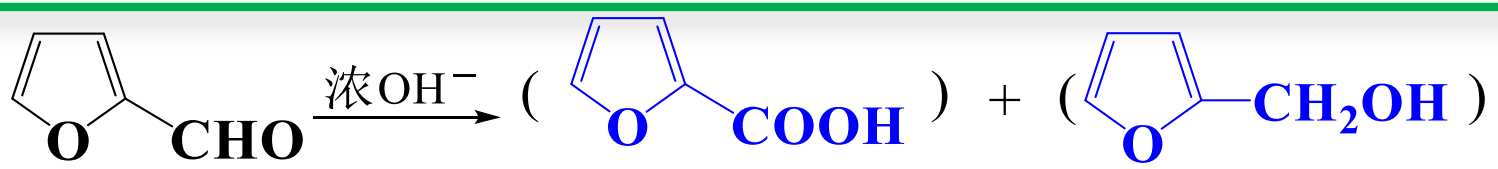
(9). β-吲哚乙酸

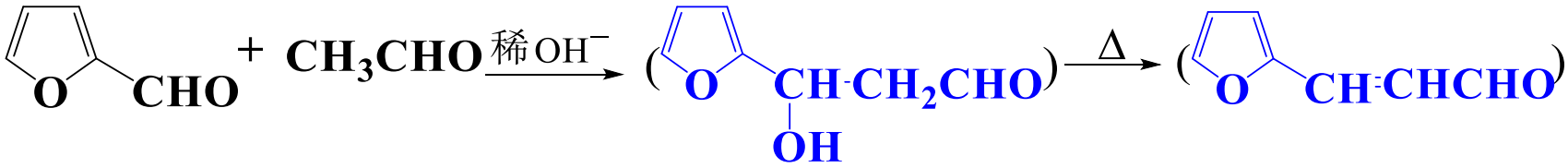


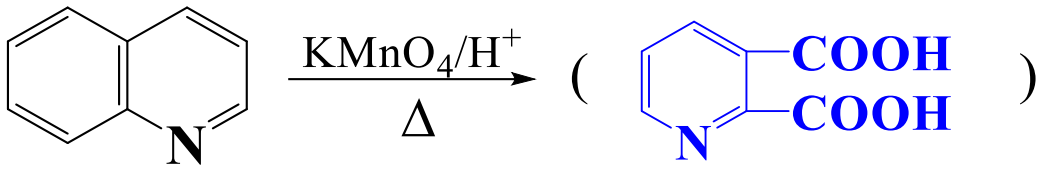
(10). 糠醛

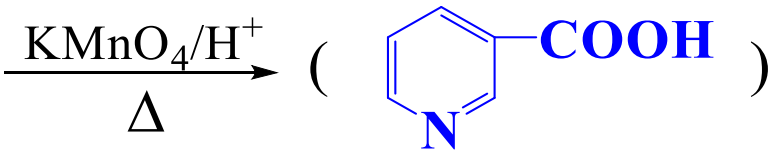


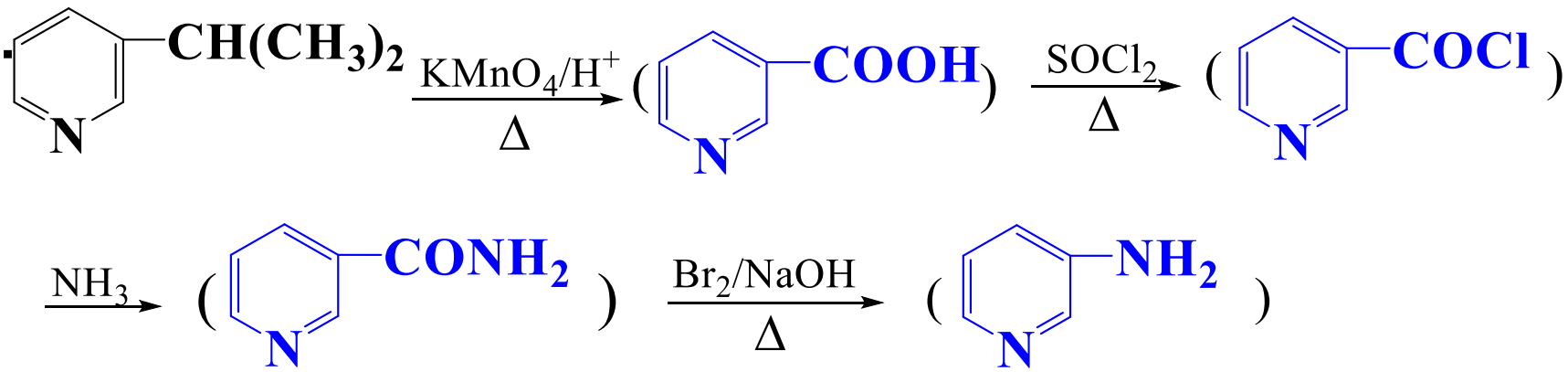
2. 完成下列反应式

(1). 

(2). 

(3). 

(4). 烟碱 

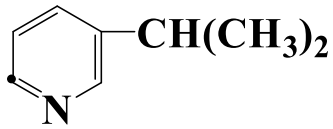
(5). 

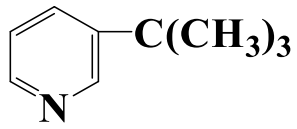
3. 选择或填空

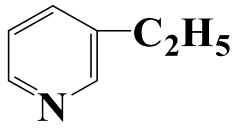
(1). 下列化合物中碱性最弱的是（ C）。

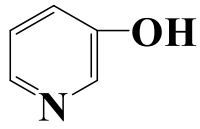
A. 氨　　B. 二甲胺　　C. 吡咯　　D. 吡啶

(2). 下列化合物中不能被$KMnO_4/H^+$氧化的是（ B ）

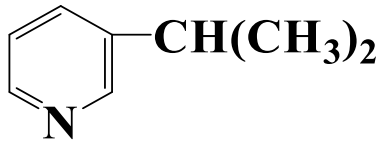
A. 

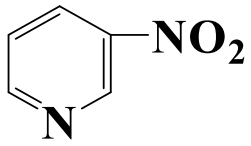
B. 

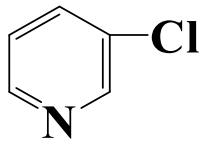
C. 

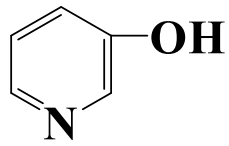
D. 

(3). 下列化合物发生硝化反应速度最快的是（ D ）

A. 

B. 

C. 

D. 

(4). 吡咯在发生亲电取代时，主要取代在（ α）位，而吡啶在发生上述反应时，主要取代在（ β ）位

4. 判断题

(1). 吡咯与呋喃的蒸汽遇松木片可产生颜色。（×）

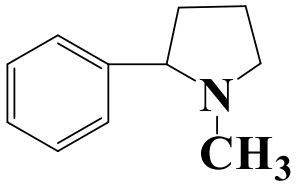
(2). 糠醛可发生银镜反应和歧化反应。（√）

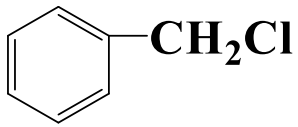
(3). 叶绿素和血红素均为吡啶的衍生物。（×）

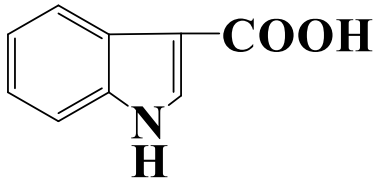
(4). 生物碱试剂可与生物碱反应生成沉淀或产生颜色变化（√）

5. 用化学方法区别下列各组化合物

(1). 烟碱　　苄氯　　β-吲哚乙酸



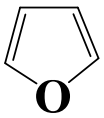


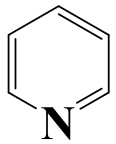


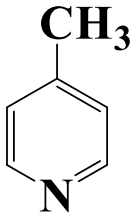
$NaHCO_3$　　(–)　　(–)　　$CO_2\uparrow$

$AgNO_3$　　(–)　　AgCl↓

(2). 呋喃　　吡啶　　4-甲基吡啶



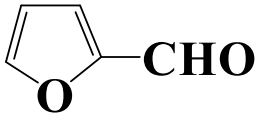


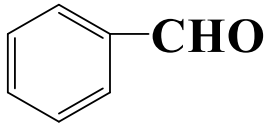


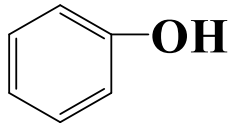
盐酸松木片反应　　显绿色　　(–)　　(–)

$KMnO_4/H^+$　　(–)　　紫红色褪去

(3). 糠醛 苯甲醛 苯酚





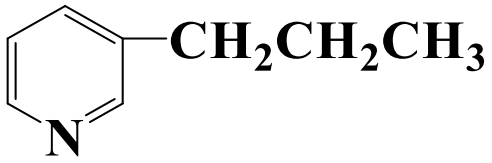


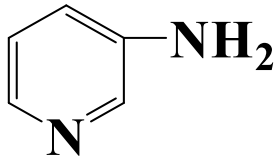
$FeCl_3$ (-) (-) 紫色

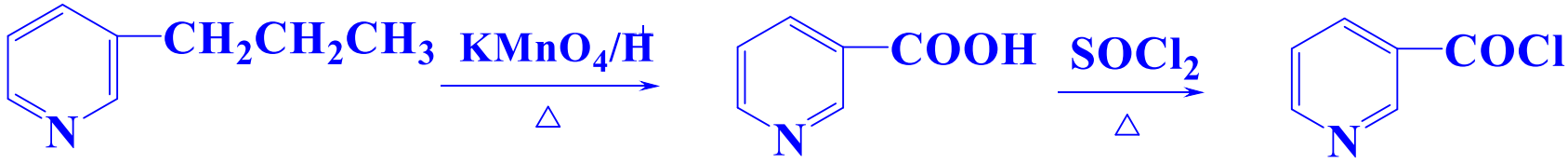
苯胺/盐酸 深红色 (-)

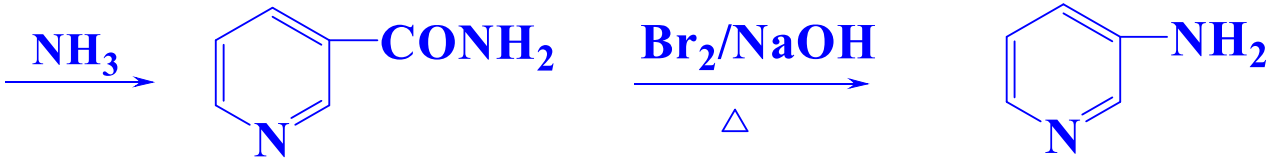
6. 合成下列化合物

(1).

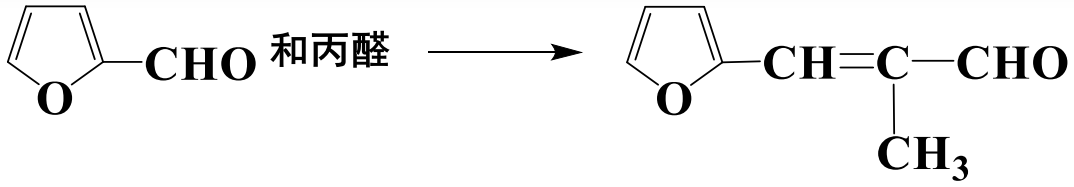


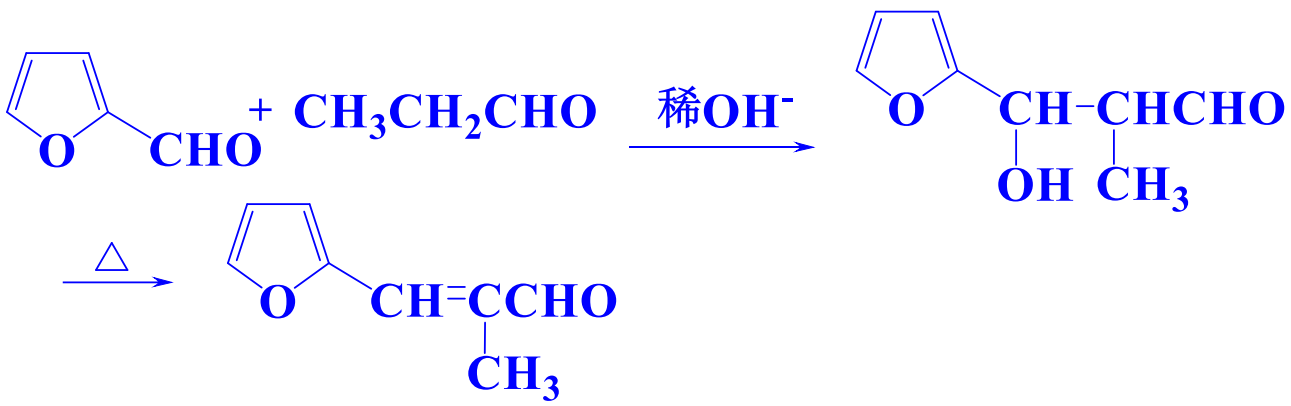




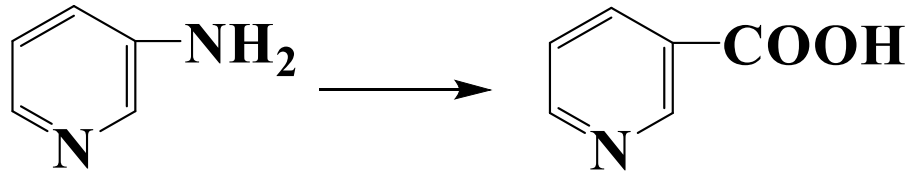


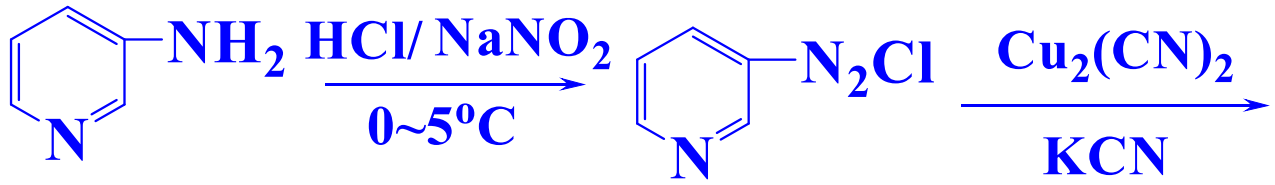
(2).

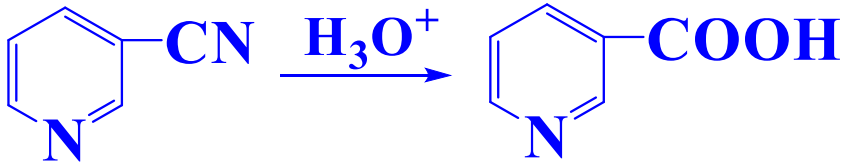

CHO
和丙醛
O
CH=C—CHO
CH3

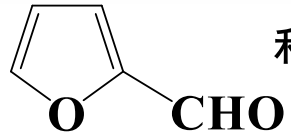
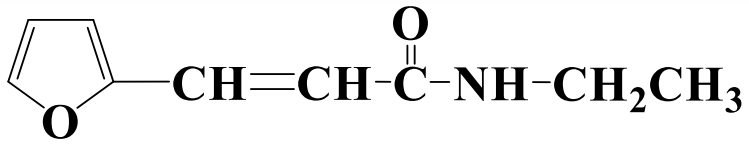

+ CH3CH2CHO
稀OH-
O
CHO
CH—CHCHO
OH CH3
△
CH=CCHO
CH3

(3).


NH2
COOH
N

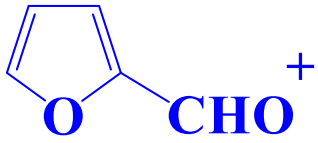
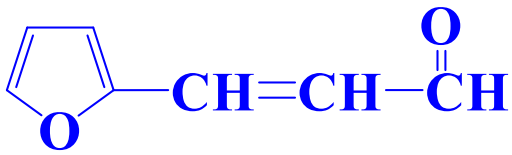

NH2
HCl/ NaNO2
0~5°C
N2Cl
Cu2(CN)2
KCN
N

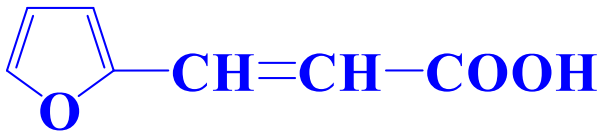
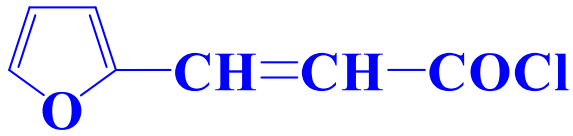

CN
H3O+
COOH
N

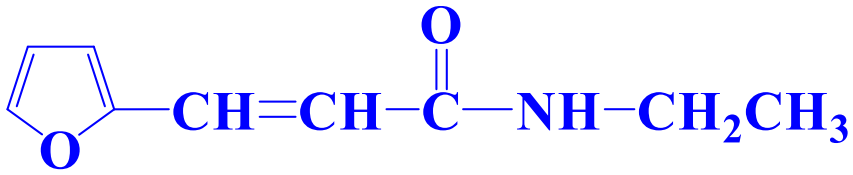
(4).  和乙醇 

$$CH_3CH_2OH \xrightarrow{Cu\ /\ 325°C} CH_3CHO$$

$$CH_3CH_2OH \xrightarrow[H^+]{HBr} CH_3CH_2Br \xrightarrow{NH_3} CH_3CH_2NH_2$$

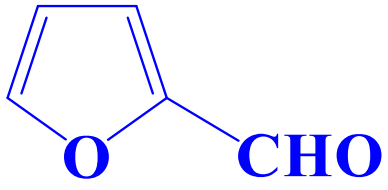
 $+\ CH_3CHO \xrightarrow{稀\ OH^-}$  $\xrightarrow{斐林试剂}$

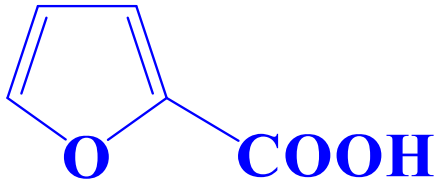
 $\xrightarrow{SOCl_2}$ 

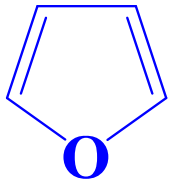
$\xrightarrow{CH_3CH_2NH_2}$ 

7. 推断题

某杂环化合物A，分子式为$C_5H_4O_2$， 经氧化后生成分子式为$C_5H_4O_3$的羧酸B，羧基与杂原子相邻，该羧酸B的钠盐与碱石灰一起加强热，则转变为分子式为C_4H_4O的化合物C；C不与金属钠作用，也不具有醛酮的性质，试写出A、B、C的结构式

A: 

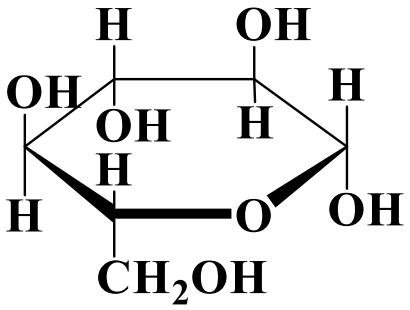
B: 

C: 

第12章 糖类化合物

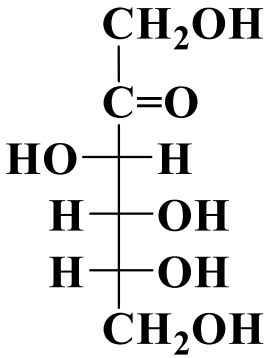
1. 命名或写结构式

(1).



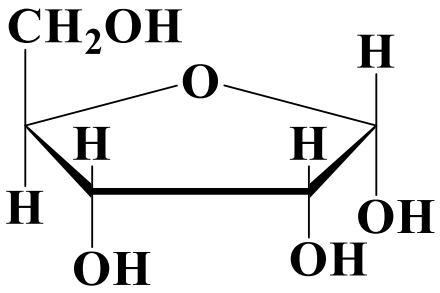
β-D-葡萄糖

(2).



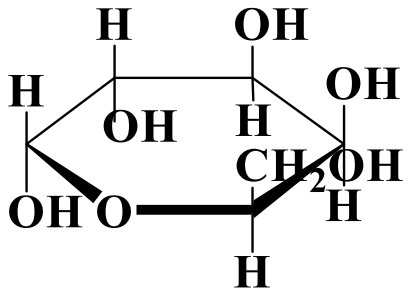
D-果糖

(3).



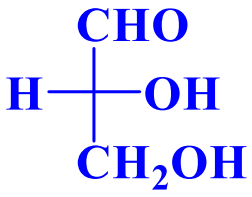
α-D-呋喃核糖

(4).

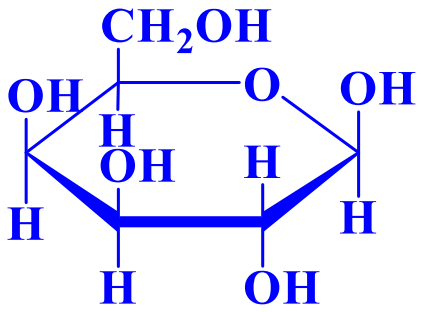


α-D-半乳糖

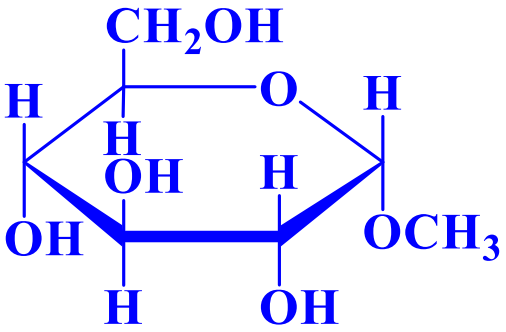
(5). D−甘油醛



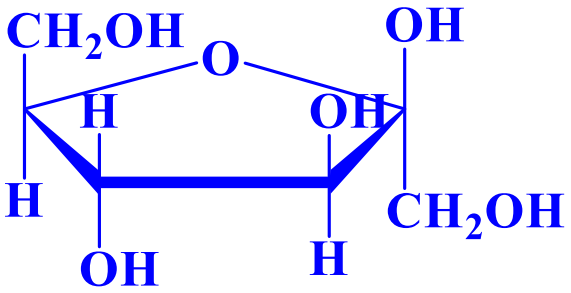
(6). β−D−半乳糖



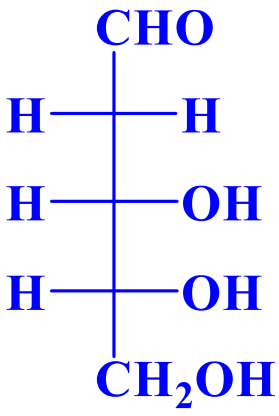
(7). 甲基−α−D−吡喃葡萄糖苷



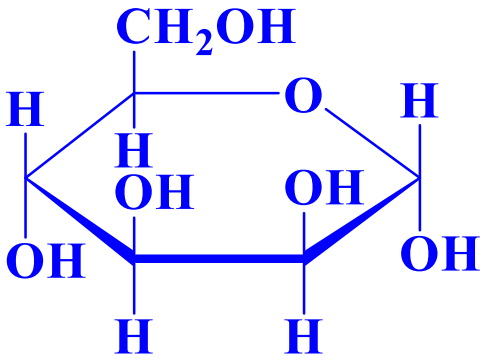
(8). β−D−呋喃果糖



(9). D−2−脱氧核糖

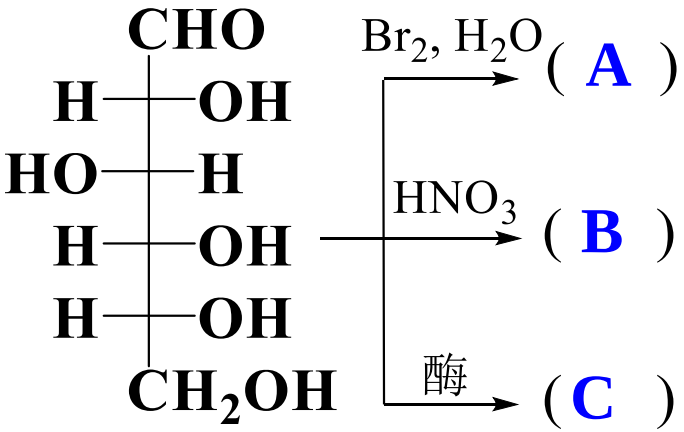


(10). α−D−甘露糖

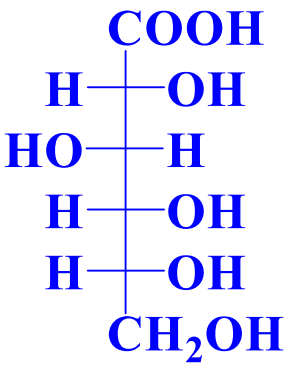


2. 完成下列反应式

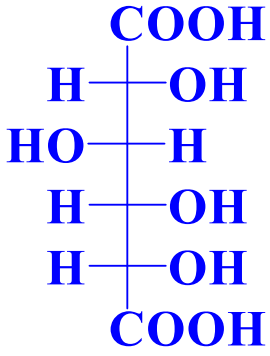
(1).



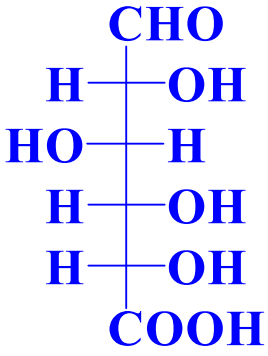
A



B

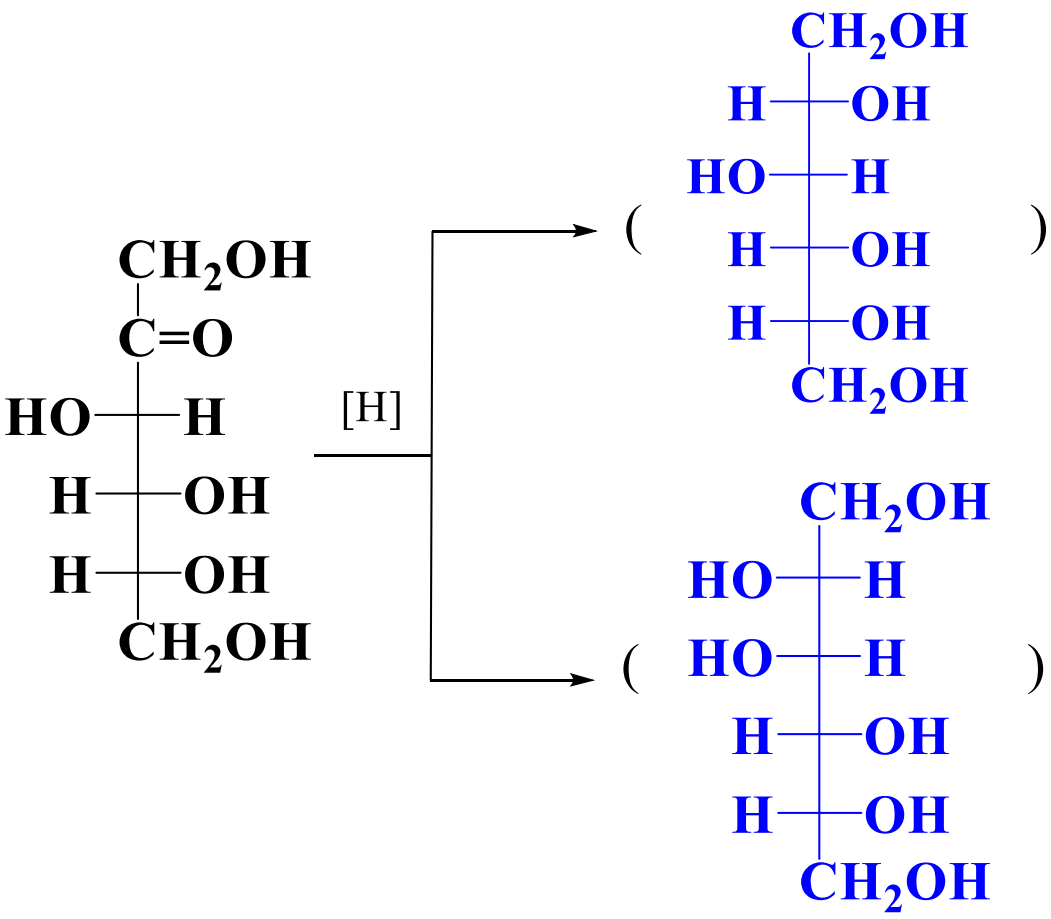


C



(2). 蔗糖 + H_2O $\xrightarrow{\text{稀酸或转化酶}}$ (D-果糖) + (D-葡萄糖)

(3).



3. 选择或填空

(1). 下列化合物不是还原性糖的是（ D ）

A. 葡萄糖　B. 麦芽糖　C . 果糖　D. 蔗糖

(2). 下列糖类化合物不属于己醛糖的是（ D）

A. 甘露糖　B. 半乳糖　C. 葡萄糖　D. 核糖

(3). 下列糖不能被溴水氧化的是（ A ）

A. 果糖　B. 甘露糖　C. 葡萄糖　D. 半乳糖

(4). α-D-葡萄糖的比旋光度为（ +112° ），β-D-葡糖糖的比旋光度为（ +19° ），将任一种配成水溶液后放置一段时间，其水溶液的比旋光度为（ +53° ）。

4. 判断题

(1). 糖苷在酸性溶液中稳定，在碱性溶液中水解。 （×）

(2). 纤维素结构稳定，在任何条件下都不能水解。（×）

(3). 海藻糖是非还原性糖, 水解后也不能还原费林试剂。（×）

(4). 单糖都有旋光性，都有变旋现象。 （×）

(5). 葡萄糖是醛糖，有还原性，而果糖是酮糖，没有还原性。（×）

5. 用化学方法区别下列各组化合物

(1). 己醛　　葡萄糖　　果糖

	己醛	葡萄糖	果糖
莫力许反应	(-)	紫色	紫色
西列瓦诺夫反应		(-)	红色

(2). 果糖　　甘露糖　　核糖

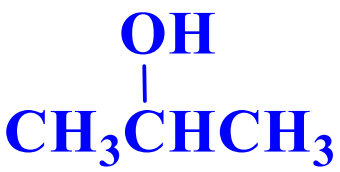
	果糖	甘露糖	核糖
皮阿尔反应	(-)	(-)	绿色
西列瓦诺夫反应	红色	(-)	

(3). 脱氧核糖 葡萄糖 蔗糖

	脱氧核糖	葡萄糖	蔗糖
狄斯克 反应	蓝色	(-)	(-)
银镜反应		Ag↓	(-)

(4). 淀粉 甲基-α-D-吡喃葡萄糖苷 麦芽糖

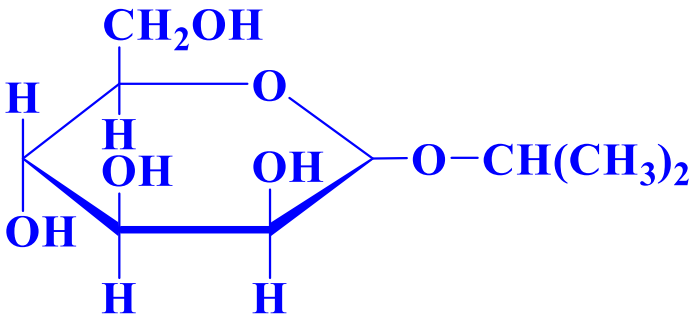
	淀粉	甲基-α-D-葡萄糖苷	麦芽糖
I_2	蓝色	(-)	(-)
银镜反应		(-)	Ag^+



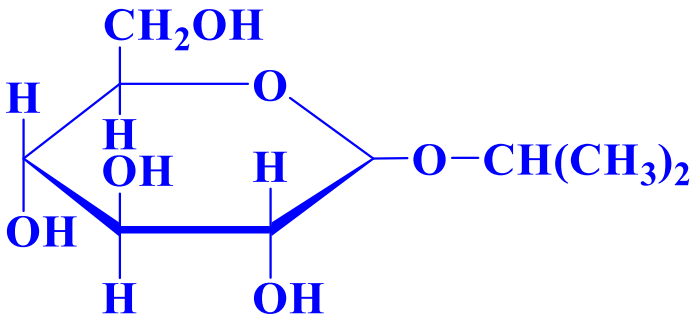
6. 推断题

(1). 化合物A($C_9H_{18}O_6$)无还原性，经水解生成化合物B和C。B($C_6H_{12}O_6$)有还原性，可被溴水氧化，与D-葡萄糖生成相同的脎。C(C_3H_8O)可发生碘仿反应。试写出A、B、C的结构式。

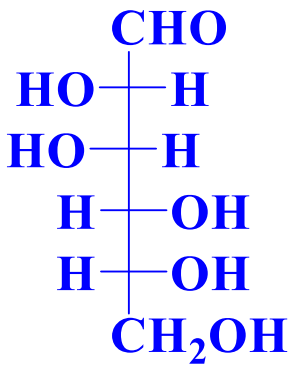
A.



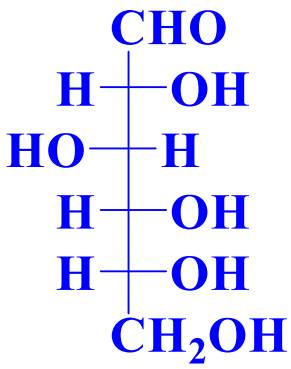
或



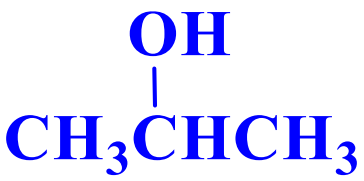
B.



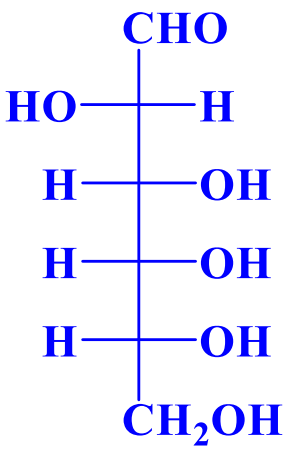
或



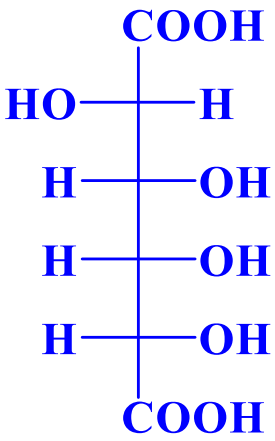
C.



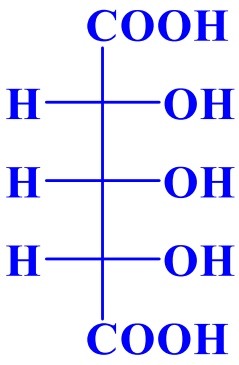
(2). 某D-型己醛糖A氧化得旋光糖二酸B，将A降解为戊醛糖后再氧化得不旋光糖二酸C。与A具有相同糖脎的另一己醛糖D氧化后得不旋光的糖二酸E。试推测A、B、C、D、E的结构



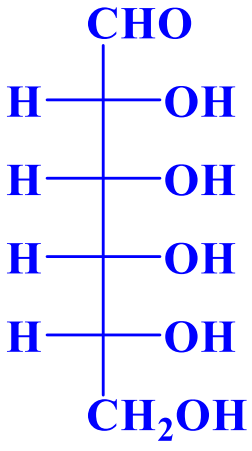
A



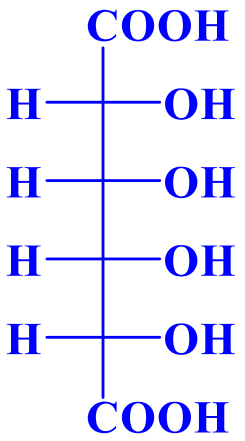
B



C



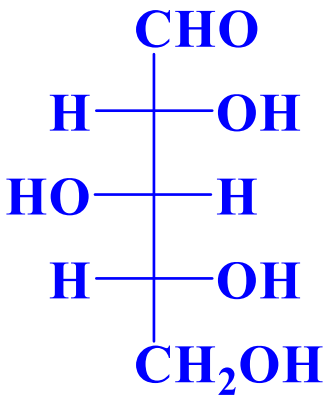
D



E

(3). 有A、B两个D-戊糖，分子式均为$C_5H_{10}O_5$，经实验得知，①将A、B分别进行西列瓦诺夫实验，B很快显红色，A则不能；②分别与苯肼试剂作用，生成两个互为差向异构体的糖脎；③对B进行C_3构型测定为R型；④将A用HNO_3氧化得到内消旋体。试写出A和B的费歇尔投影式及有关反应式

A.



B.

